

Create & Connect to A Windows Virtual Machine (EC2 Instance)

[Edition 22]

[Last Update 231210]

Scenario Overview:

In this hands-on lab, you will embark on the journey of creating and connecting to a Windows EC2 instance on Amazon Web Services (AWS). This practical exercise is designed to familiarize you with the fundamental steps of setting up a virtual Windows server in the cloud and establishing a secure connection for remote access.

This lab serves as a foundational exploration into the creation and connection to a Windows EC2 instance, providing essential skills for cloud-based computing and fostering a deeper understanding of AWS infrastructure and virtualization.

For any issues/help contact in Training WhatsApp Group or Support Ticketing System

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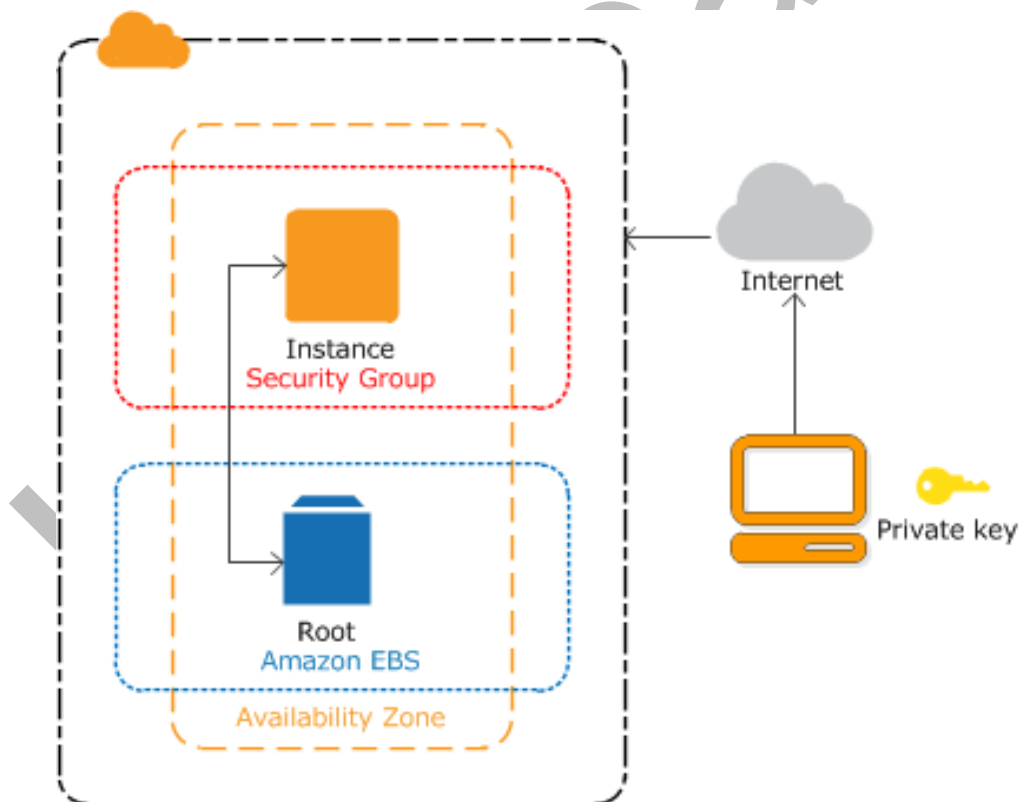
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1 INTRODUCTION

Amazon EC2 presents a true virtual computing environment, allowing you to use web service interfaces to launch instances with a variety of operating systems, load them with your custom application environment, manage your network's access permissions, and run your image using as many or few systems as you desire.

To use Amazon EC2, you are simply:

1. Select a pre-configured, templated Amazon Machine Image (AMI) to get up and running immediately, or create an AMI containing your applications, libraries, data, and associated configuration settings.
2. Configure security and network access on your Amazon EC2 instance.
3. Choose which instance type(s) you want, then start, terminate, and monitor as many instances of your AMI as needed, using the web service APIs or the variety of management tools provided.
4. Determine whether you want to run in multiple locations, utilize static IP endpoints, or attach persistent block storage to your instances.
5. Pay only for the resources that you actually consume, like instance-hours or data transfer.



This activity guide cover steps for:

1. Creating a Windows EC2 Instance
 - a. Launching the Windows Instance
 - b. Connecting to the Windows Instance
2. Deleting/Cleanup
 - a. Terminating the Windows Instance
 - b. Termination Prevention of Instance

2 OVERVIEW & KEY CONCEPTS

In this section we'll cover key concepts used in this guide, including an overview of what's covered in this guide.

2.1 What is EC2?

Amazon Elastic Compute Cloud (Amazon EC2) delivers scalable computing power on-demand in the Amazon Web Services (AWS) Cloud. Using Amazon EC2 lowers hardware expenses, allowing you to create and deploy apps quicker. Amazon EC2 allows you to deploy as many or as few virtual servers as you need, set security and networking, and manage storage. You can add capacity (scale up) to address compute-intensive operations such as monthly or annual procedures or surges in website traffic. When consumption falls, you can lower capacity.

The EC2 instance is located in a Region Availability Zone. A security group, which is a virtual firewall that regulates incoming and outgoing traffic, protects the EC2 instance. A private key is kept on the local computer, while a public key is kept on the instance. To establish the user's identity, both keys are supplied as a key pair. The instance in this case is backed up using an Amazon EBS volume. An internet gateway connects the VPC to the internet.

2.2 Windows EC2 Instance

A Windows EC2 Instance refers to a virtual server running the Microsoft Windows Server operating system within Amazon Web Services' (AWS) Elastic Compute Cloud (EC2). This service allows users to deploy scalable and resizable Windows-based virtual machines in the cloud. The Windows EC2 Instance is a fundamental building block in AWS, providing a reliable and flexible solution for hosting a variety of Windows-based applications, services, and workloads.

3 DOCUMENTATION LINKS

1. Amazon EC2

<https://aws.amazon.com/ec2/>

2. Features of Amazon EC2

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html>

3. Amazon EC2 FAQs

<https://aws.amazon.com/ec2/faqs/>

4. Creating a Windows Instance Using Amazon EC2

https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/EC2_GetStarted.html#ec2-launch-instance

4 PRE-REQUISITE

An AWS Account (Free or Paid).

Kindly refer to the ****Create AWS Free Trial Account**** Activity guide.

K21Academy

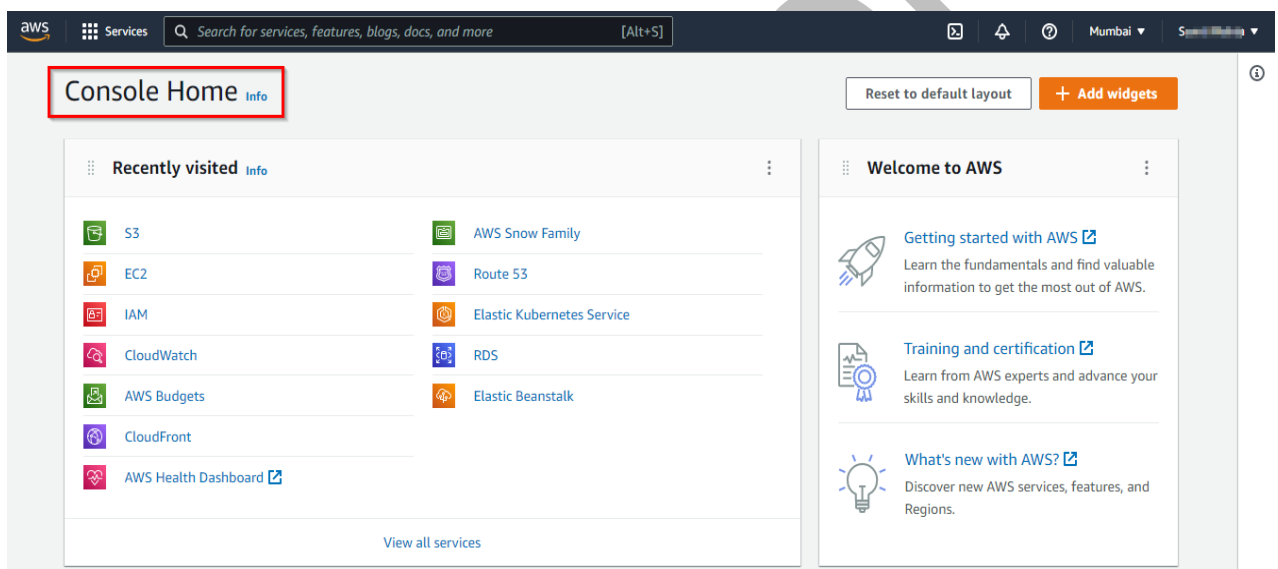
5 CREATING A WINDOWS EC2 INSTANCE

Creating a Windows EC2 instance refers to the process of setting up a virtual server on the Amazon Web Services (AWS) cloud platform that runs the Microsoft Windows operating system. EC2 stands for Elastic Compute Cloud, which is a service provided by AWS that enables customers to launch and manage virtual servers in the cloud.

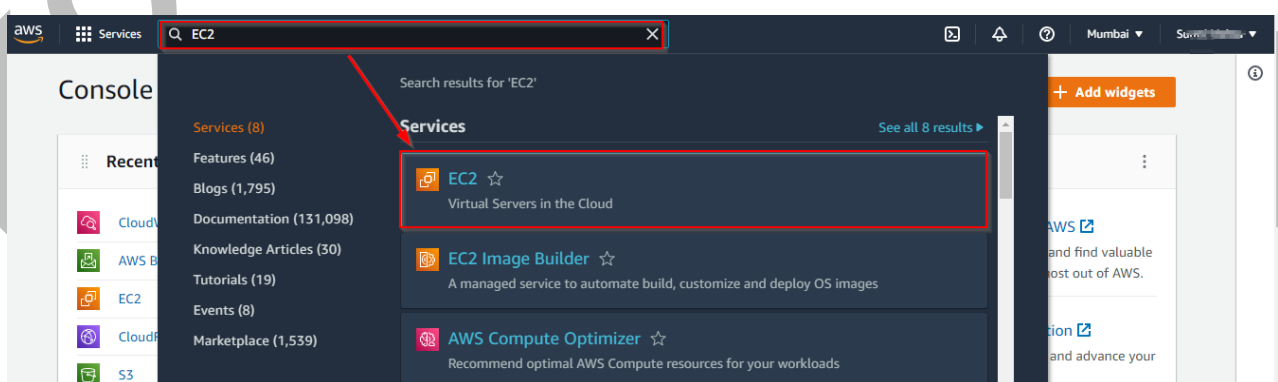
5.1 Launching the Windows Instance

Note: In this section we will be creating a Windows Server 2022 Base

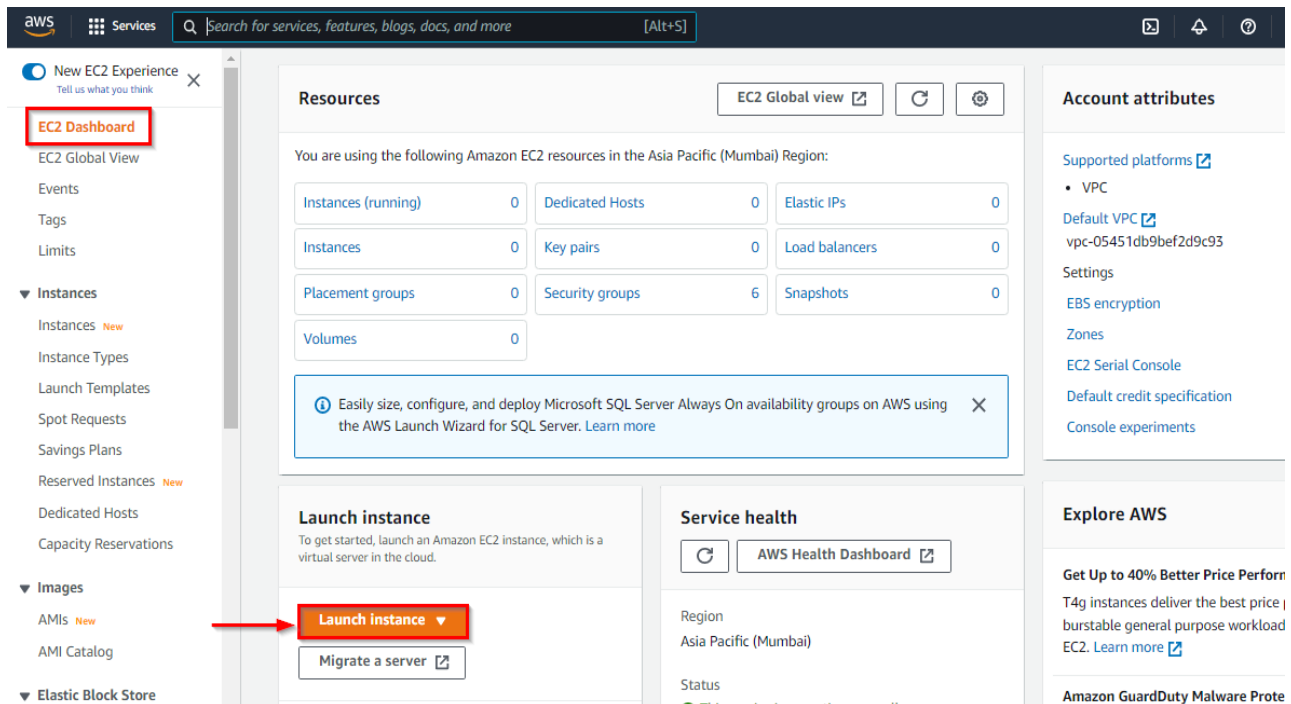
1. Log in to your AWS console.



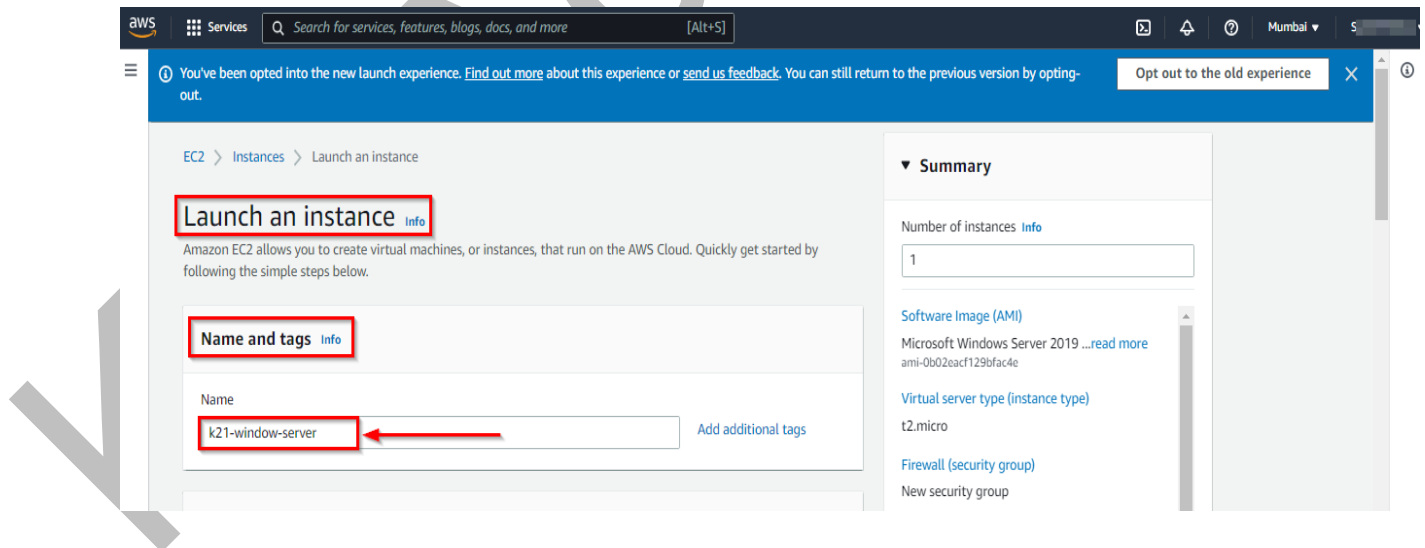
2. Search for EC2 in the search bar and click on it.



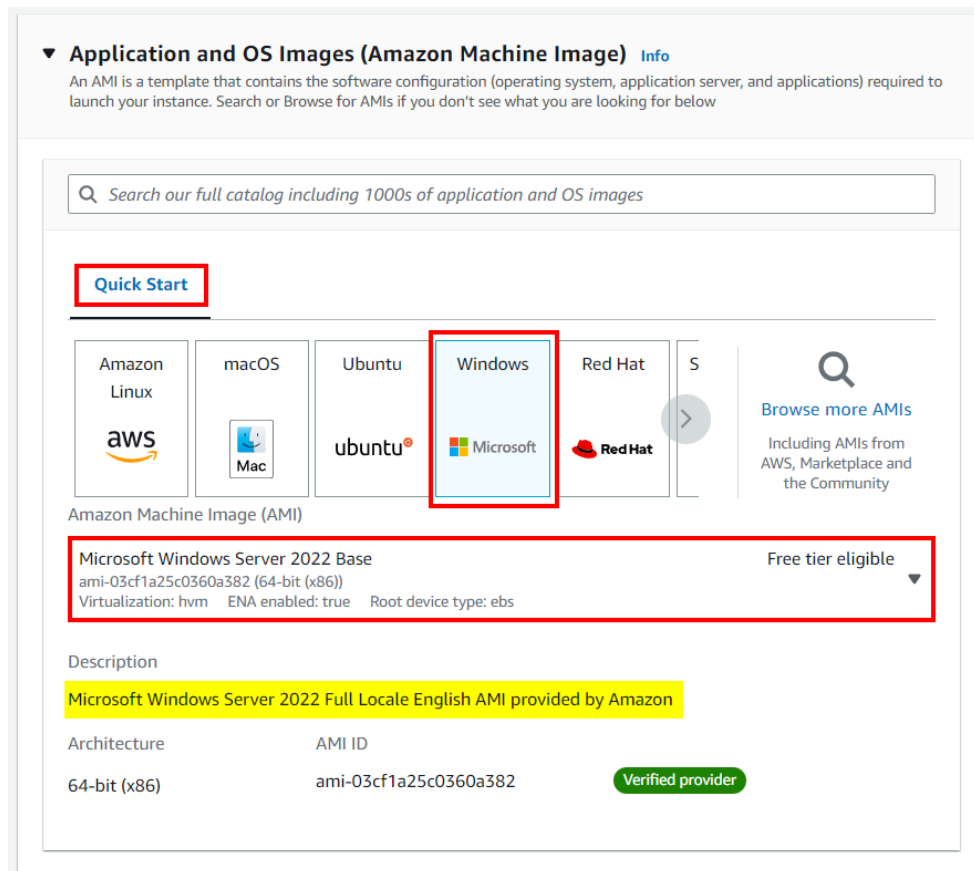
3. Once you are at the EC2 Dashboard, click on **Launch** to launch an instance.



4. In the Name and Tags step you can add tags to an instance, here tags help you to enable categorize AWS resources in different ways, for example, by owner, environment, or purpose.



5. In Select **Windows** under QuickStart and Select **Microsoft Windows Server 2022 Base AMI**. You can also select other AMI as per your need but here we are launching a Windows Server, so we have to select the Windows Server 2022 Base AMI.



Note: Make sure you choose same AMI “**Microsoft Windows Server 2022 Base**” so that you get that same output as we are getting in this lab.

6. For Select the **t2.micro** instance type, if you want you may select another instance type but they are chargeable so we choose the t2.micro instance type which is eligible for the free tier and limited resources.

▼ Instance type [Info](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory

On-Demand Windows pricing: 0.0162 USD per Hour

On-Demand SUSE pricing: 0.0116 USD per Hour

On-Demand RHEL pricing: 0.0716 USD per Hour

On-Demand Linux pricing: 0.0116 USD per Hour

Free tier eligible

▼

Compare instance types

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

▼

Create new key pair

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

▼ Network settings [Info](#)

Edit

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Microsoft Windows Server 2022 ...[read more](#)

ami-0e38fa17744b2f6a5

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 30 GiB

Free tier: In your first year includes 750 hours of t2.micro for t3.micro in the

Cancel

Launch instance

7. Select an existing key pair or create a new one, we will Create a new one, enter the name of the Key-pair as Windows-Key and Create the Key Pair.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select

▼

Create new key pair

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

Create key pair

×

Key pairs allow you to connect to your instance securely.

Enter the name of the key pair below. When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Key pair name

window-key

The name can include upto 255 ASCII characters. It can't include leading or trailing spaces.

Private key file format

☒ .pem

For use with OpenSSH

☐ .ppk

For use with PuTTY

Cancel

Create key pair

8. Now, keep everything default and click on **Launch Instance**.

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage Info Advanced

1x 30 GiB gp2 Root volume (Not encrypted)

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

The selected AMI contains more instance store volumes than the instance allows. Only the first 0 instance store volumes from the AMI will be accessible from the instance

0 x File systems Edit

Advanced details Info

Summary

Number of instances Info
1

Software Image (AMI)
Microsoft Windows Server 2022 ...read more
ami-0e38fa17744b2f6a5

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

Free tier: In your first year includes 750 hours of t2.micro for t3.micro in the

Cancel Launch instance

9. Now Click on **View all Instances**.

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-062c260586ec8a81f)
Launch log

Next Steps

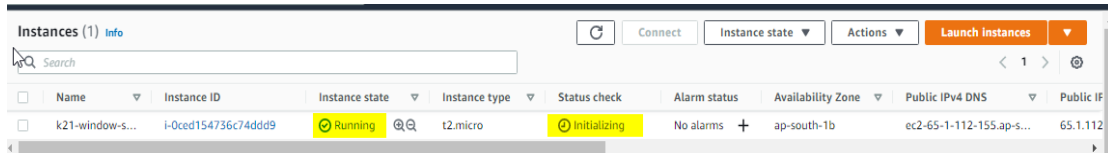
Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
Create billing alerts

Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more

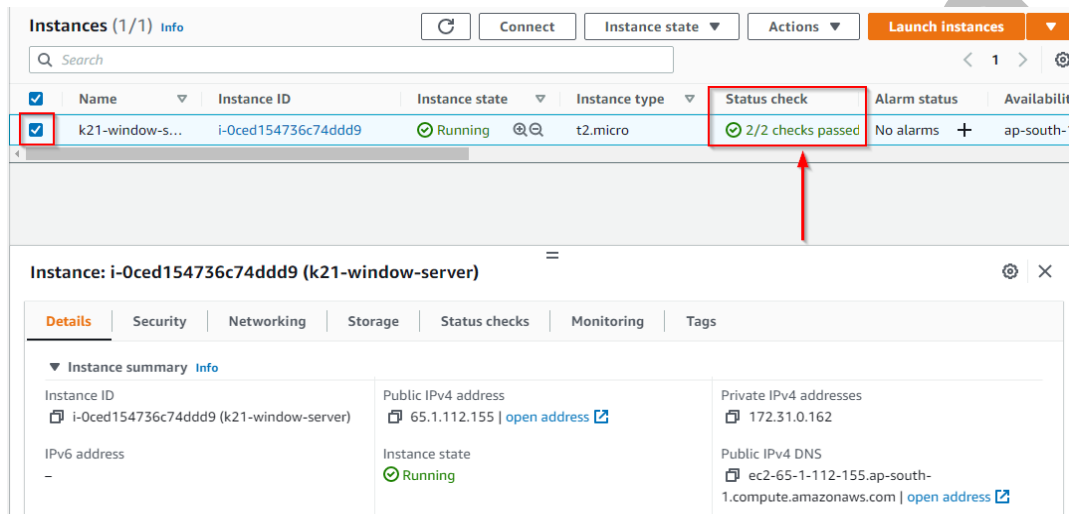
View all instances

10. Here, you shall see your instance is launching and the Status check is Initializing, wait for some time.



	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IP
☐	k21-window-s...	i-0ced154736c74ddd9	Running	t2.micro	Initializing	No alarms	ap-south-1b	ec2-65-1-112-155.ap-s...	65.1.112

11. Refresh and you shall see your instance are Up and Running, and the Status check has changed to 2/2 checks.



	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☒	k21-window-s...	i-0ced154736c74ddd9	Running	t2.micro	2/2 checks passed	No alarms	ap-south-

Instance: i-0ced154736c74ddd9 (k21-window-server)

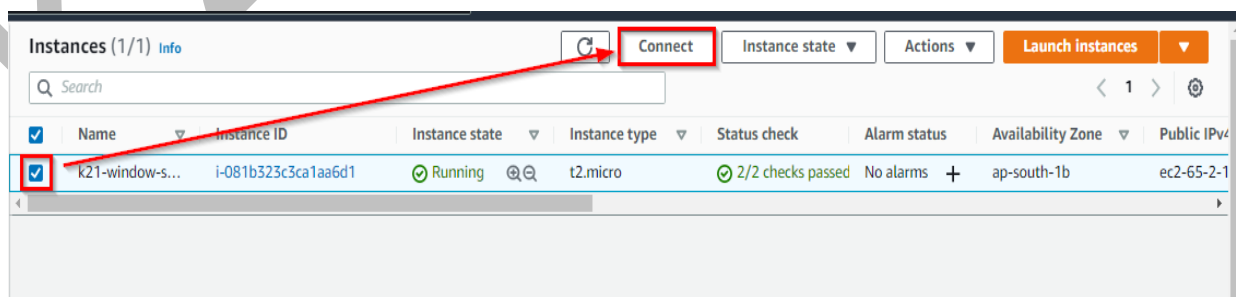
Details	Security	Networking	Storage	Status checks	Monitoring	Tags
Instance summary Info Instance ID i-0ced154736c74ddd9 (k21-window-server) Public IPv4 address 65.1.112.155 open address Private IPv4 addresses 172.31.0.162 IPv6 address - Instance state Running Public IPv4 DNS ec2-65-1-112-155.ap-south-1.compute.amazonaws.com open address						

Now we have successfully created our Windows EC2 Instance.

5.2 Connecting to the Windows Instance

Note: In this section we will be connecting to our Windows Instance using Remote Desktop Connection.

1. Move to your **Instance** dashboard and select the instance we just created. Click on **Connect**.



	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☒	k21-window-s...	i-081b323c3ca1aa6d1	Running	t2.micro	2/2 checks passed	No alarms	ap-south-1b	ec2-65-2-1

- Here we have to select the **RDP (Remote desktop protocol)** Client and then Download the RDP File and save it somewhere safe then, and we need a password to access the RDP file, so click on Get Password.

EC2 > Instances > i-04ceae13ea444d15e > Connect to instance

Connect to instance [Info](#)

Connect to your instance i-04ceae13ea444d15e (k21-windows-server) using any of these options

Session Manager **RDP client** EC2 serial console

Instance ID
i-04ceae13ea444d15e (k21-windows-server)

Connection Type

Connect using RDP client

Download a file to use with your RDP client and retrieve your password.

Connect using Fleet Manager

To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

Download remote desktop file 1

When prompted, connect to your instance using the following details:

Public DNS
ec2-100-25-169-135.compute-1.amazonaws.com

User name
Administrator

Password **Get password** 2

- Once you click on the **Get Password**, click on **Upload private key file** and select the **key pair** we just created at the time of instance launching.

EC2 > Instances > i-04ceae13ea444d15e > Get Windows password

Get Windows password [Info](#)

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID
i-04ceae13ea444d15e (k21-windows-server)

Key pair associated with this instance
windows-key

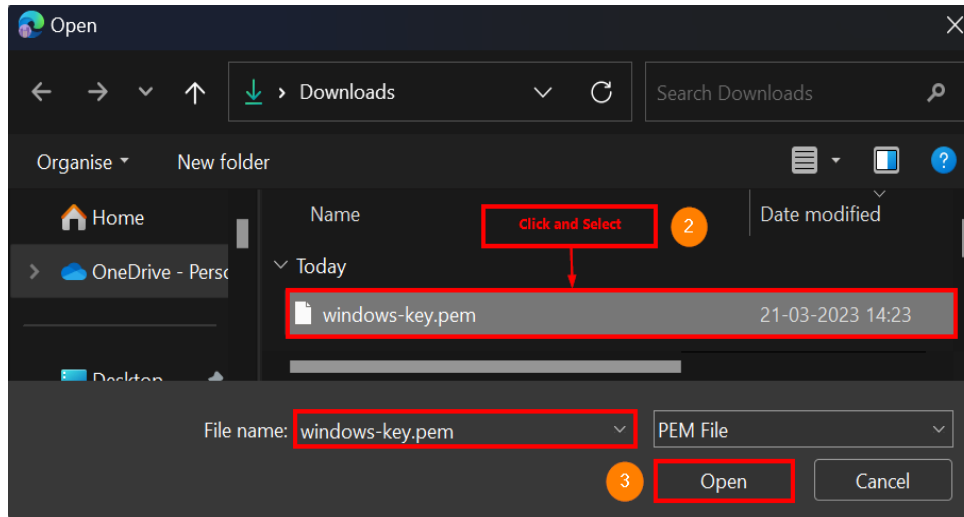
Private key
Either upload your private key file or copy and paste its contents into the field below.

Upload private key file 1

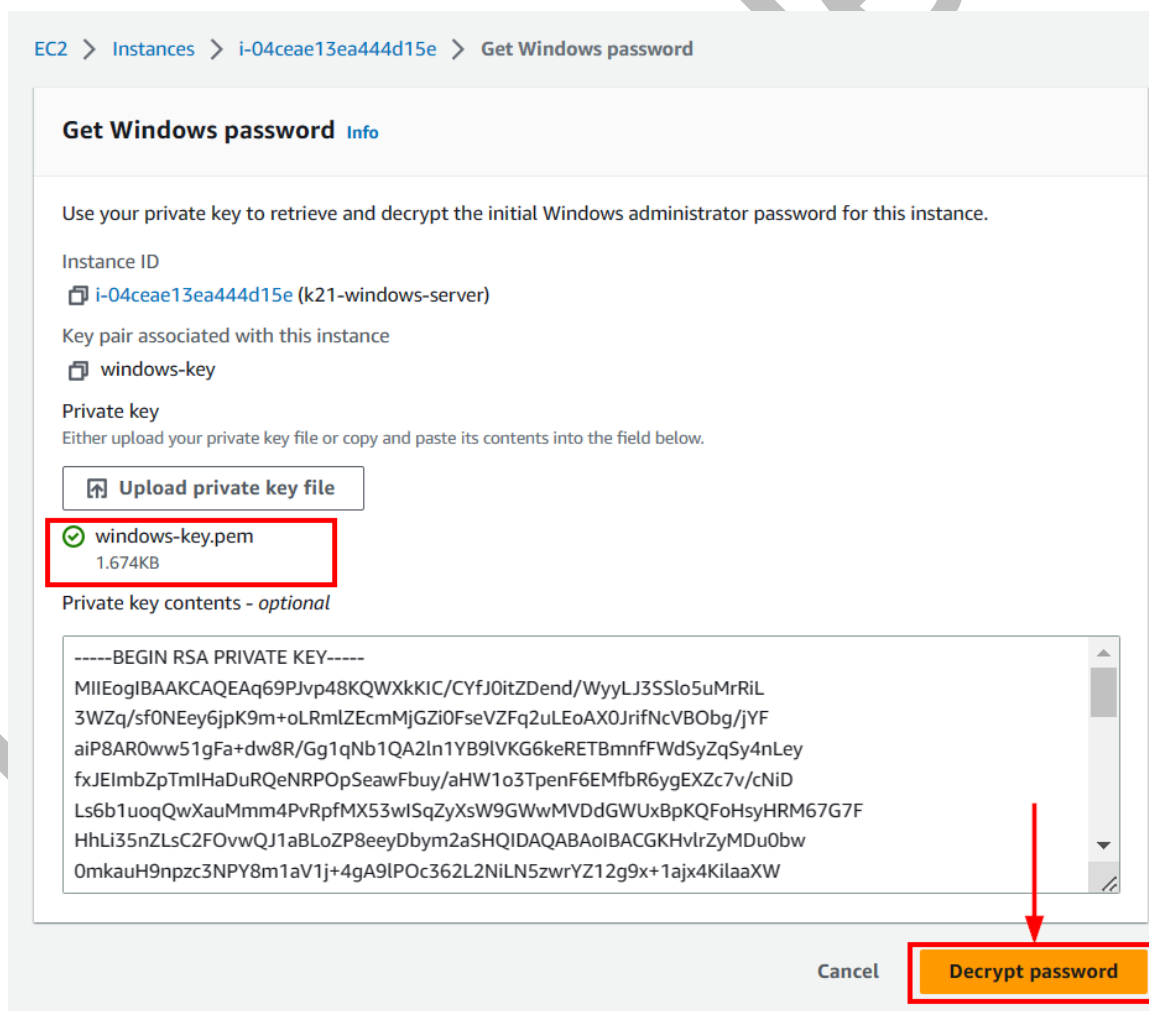
Private key contents - optional

Private key contents

Cancel Decrypt password



4. Now click on **Decrypt Password**.



5. After this you will get the **decrypted password** for the Instance.

✓ Password Decryption Successful
The password for instance i-081b323c3ca1aa6d1 was successfully decrypted.

EC2 > Instances > i-081b323c3ca1aa6d1 > Connect to instance

Connect to instance [Info](#)

Connect to your instance i-081b323c3ca1aa6d1 (k21-window-server) using any of these options

Session Manager | **RDP client** | EC2 serial console

Instance ID
i-081b323c3ca1aa6d1 (k21-window-server)

Connection Type

☒ **Connect using RDP client**
Download a file to use with your RDP client and retrieve your password.

☐ **Connect using Fleet Manager**
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following details:

Public DNS
ec2-65-2-150-253.ap-south-1.compute.amazonaws.com

User name
Administrator

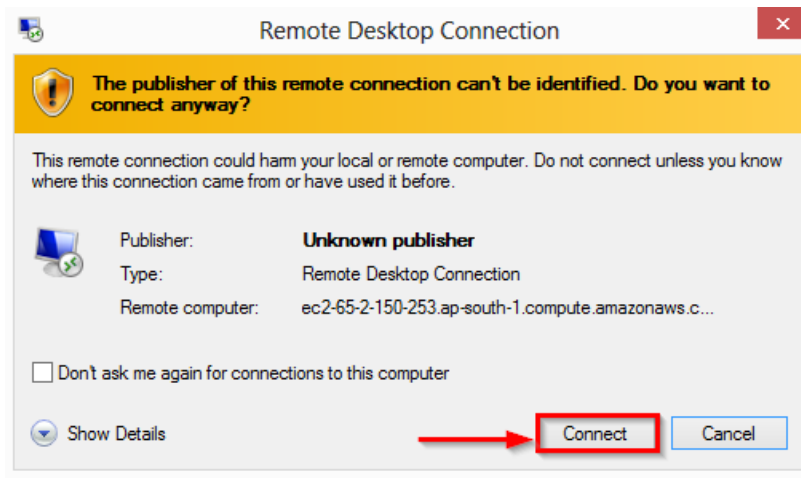
Password
FzGjn9(vqXiCo)IHnVETw=vz2NY5g(ra) ←

[If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.](#)

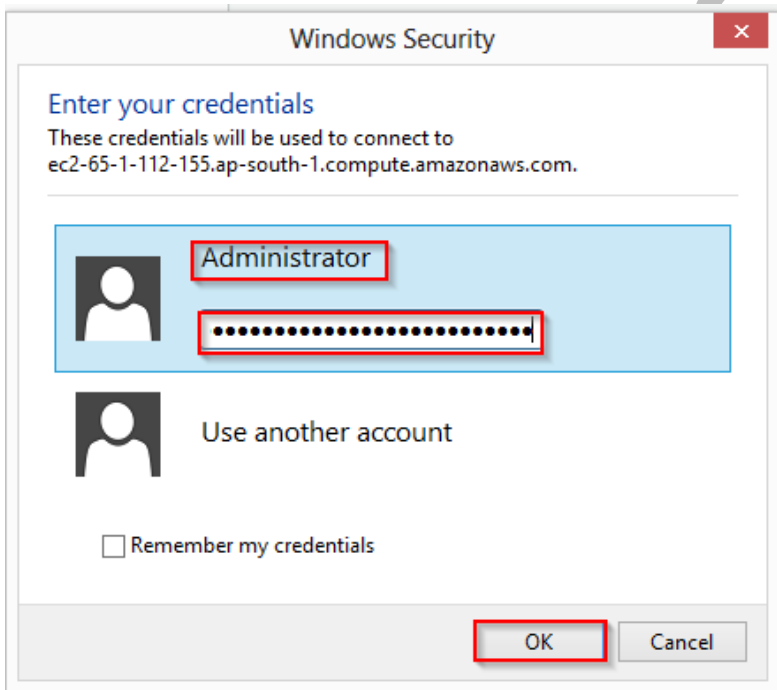
6. Now open the **Remote Desktop File** from downloads for launching the Windows instance.

Name	Date modified	Type	Size
Today (7)			
[blurred]	01-09-2022 13:31	F [blurred]	1 KB
k21-window-server	01-09-2022 13:31	Remote Desktop ...	1 KB
Windows-key.pem	01-09-2022 13:14	PEM File	2 KB

7. After opening the RDP file click on **Connect** to launch the Window instance.

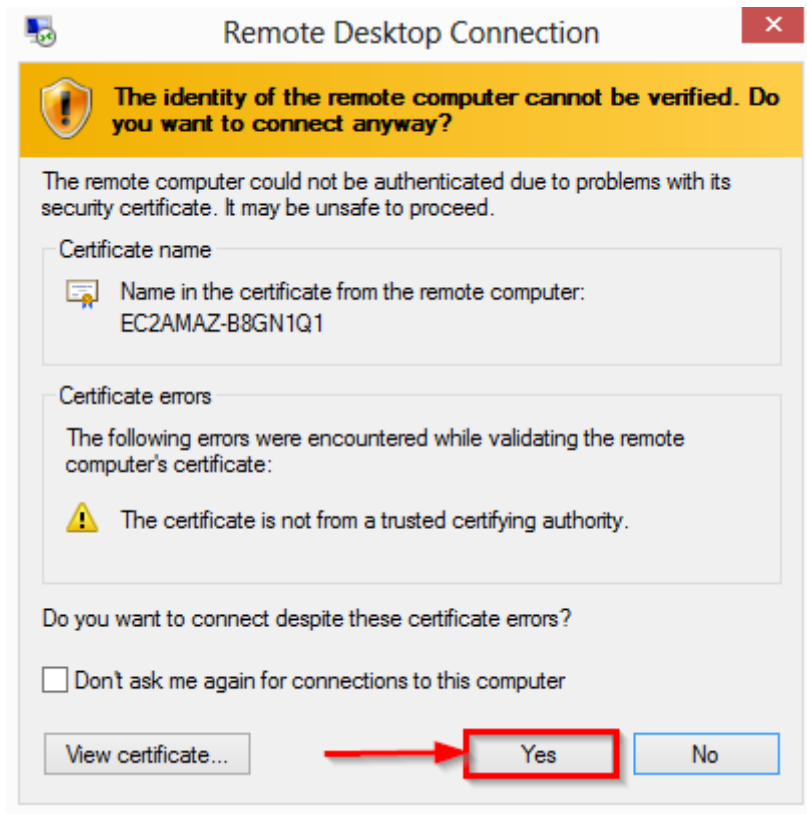


8. Here we have to provide the credentials for accessing the Instance, so we have to **Enter the Password** That we copied and click on **Ok**.

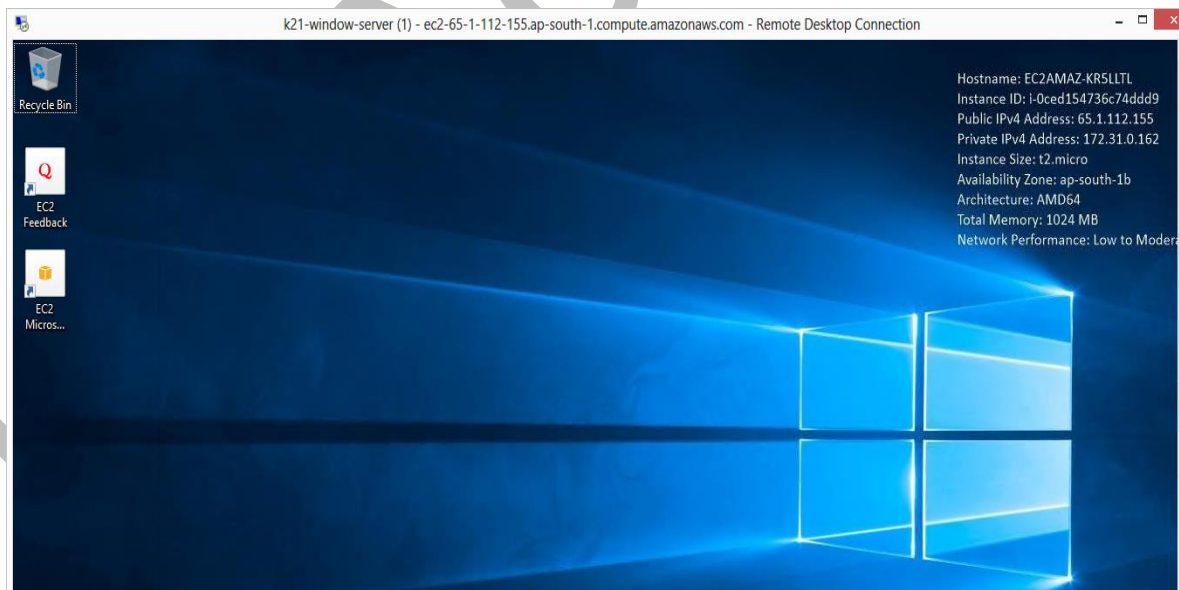


Note: If you hit error at this step by saying that “Not able to connect to RDP”, Kindly refer to the Troubleshooting Section 9.1

9. Click on **Yes** to accept the certificates.



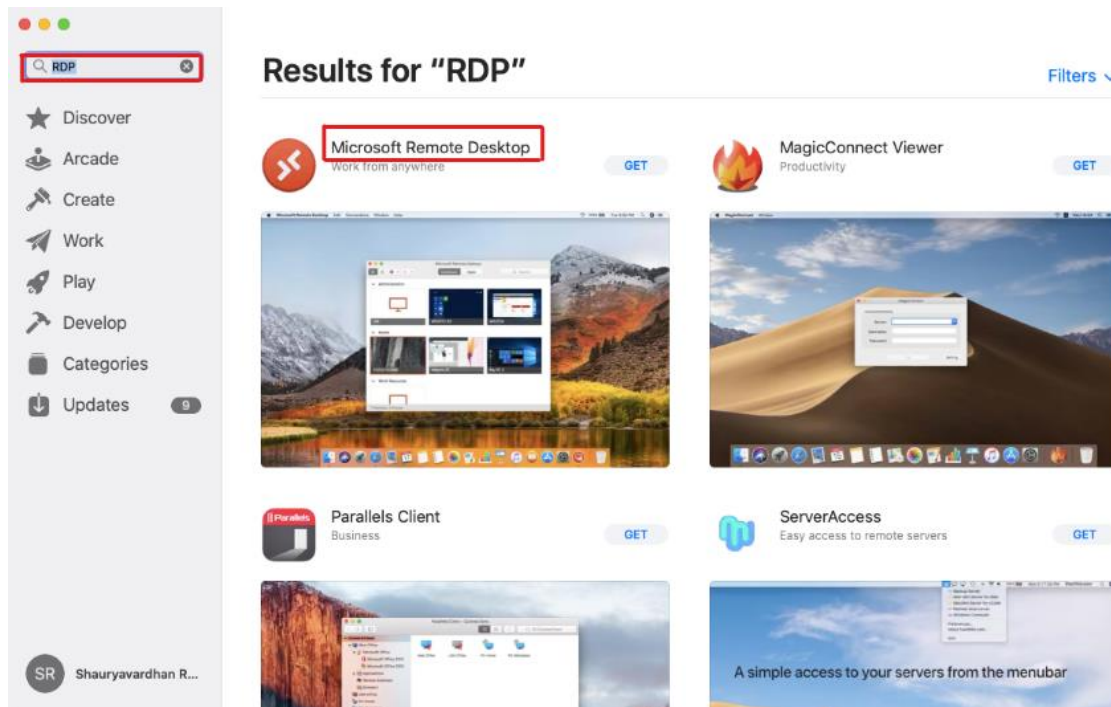
10. Now we have successfully connected to the windows instance.



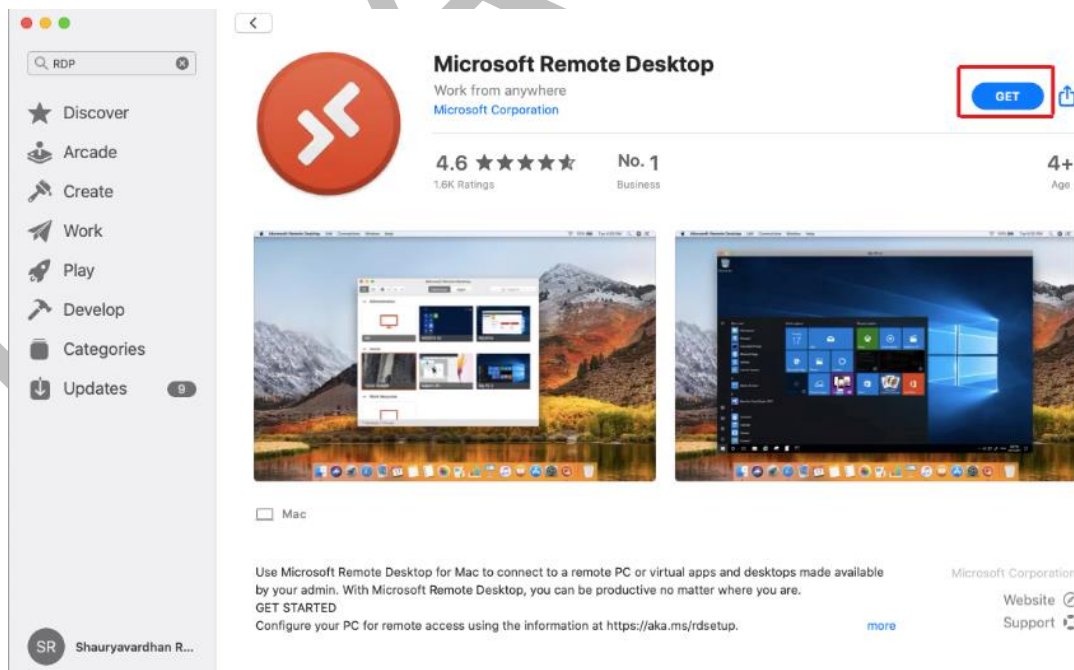
5.3 RDP On Mac

If you are using MAC, then follow below steps else you can skip this section

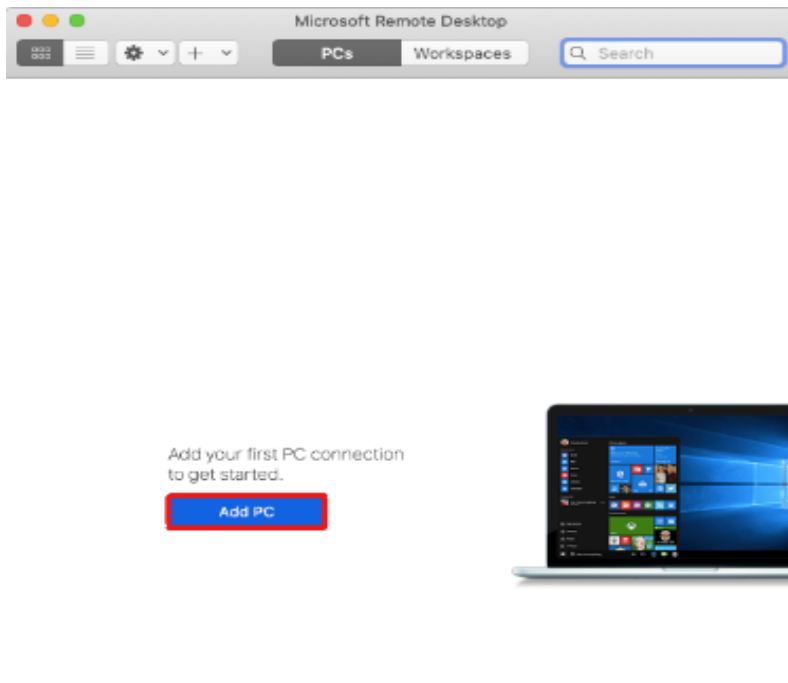
1. In the Search box, type RDP and from results choose the RDP app.



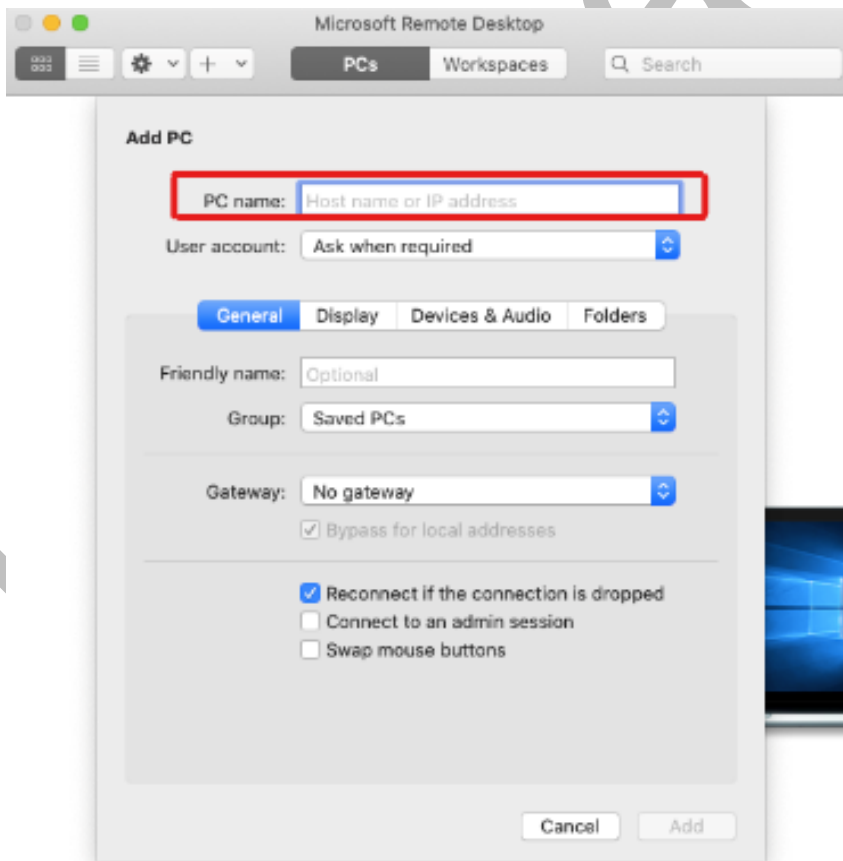
2. Click on **Get**



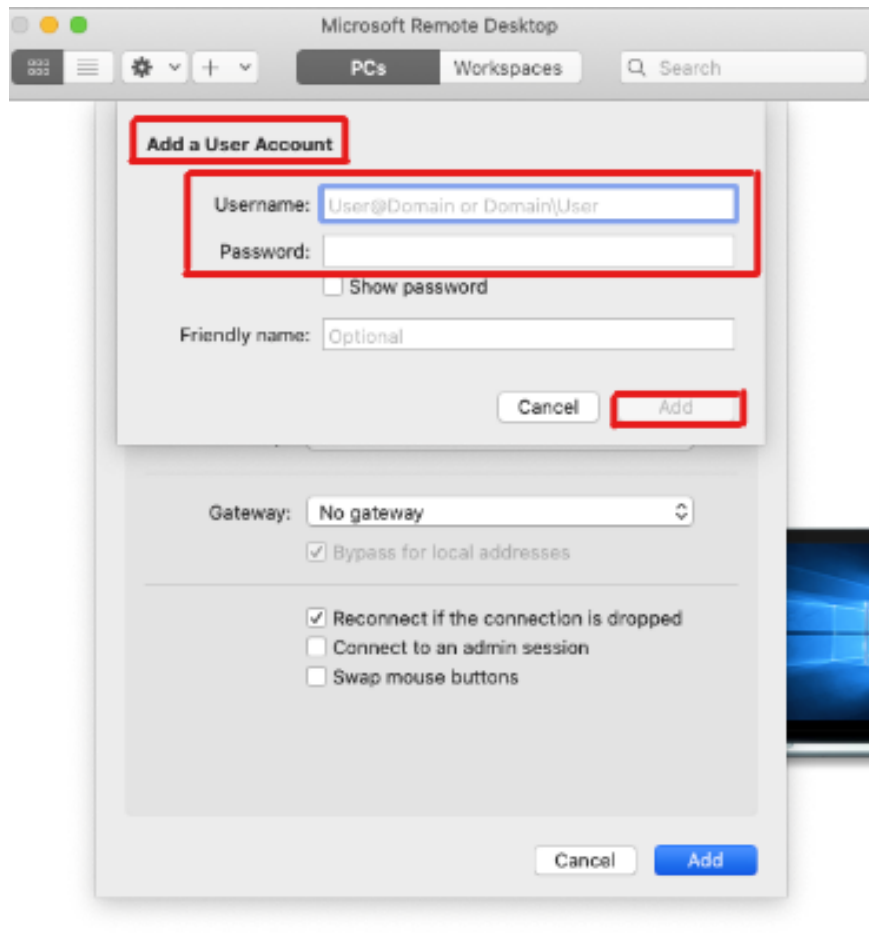
3. Open the app and click on **Add PC**.



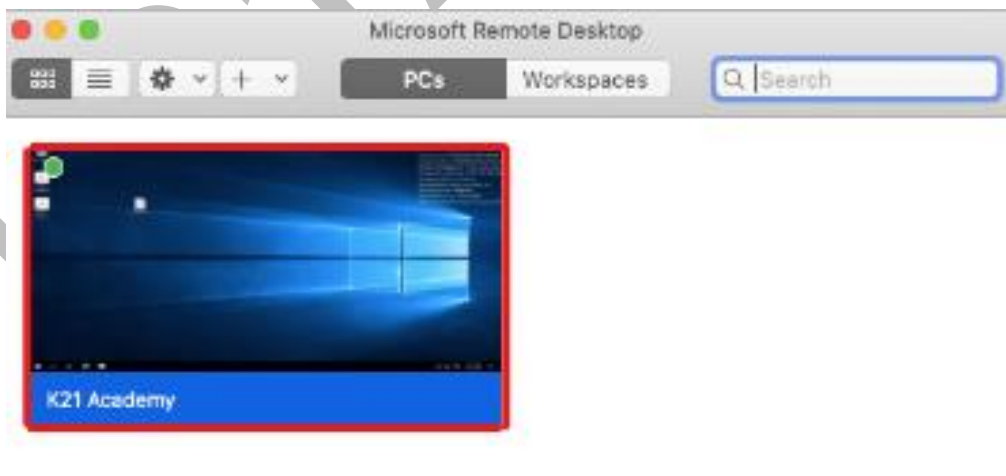
4. Give the **PC Name** (Host Name or IP Address)



5. Now Add a User Account by giving the **Username** and **Password** and click on Add.



6. Now your remote desktop is working.



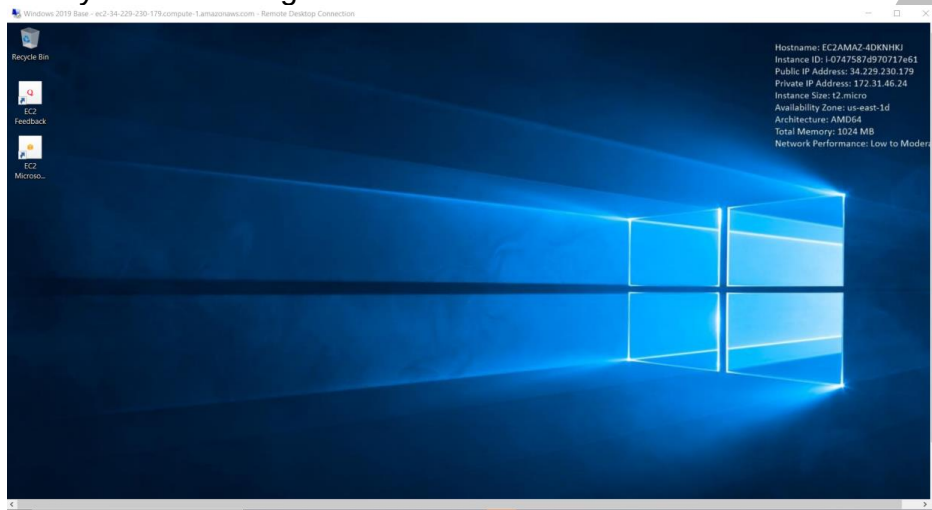
Thus, we have successfully connected to the Windows Machine on Mac.

6 RESET PASSWORD FOR WINDOWS INSTANCE

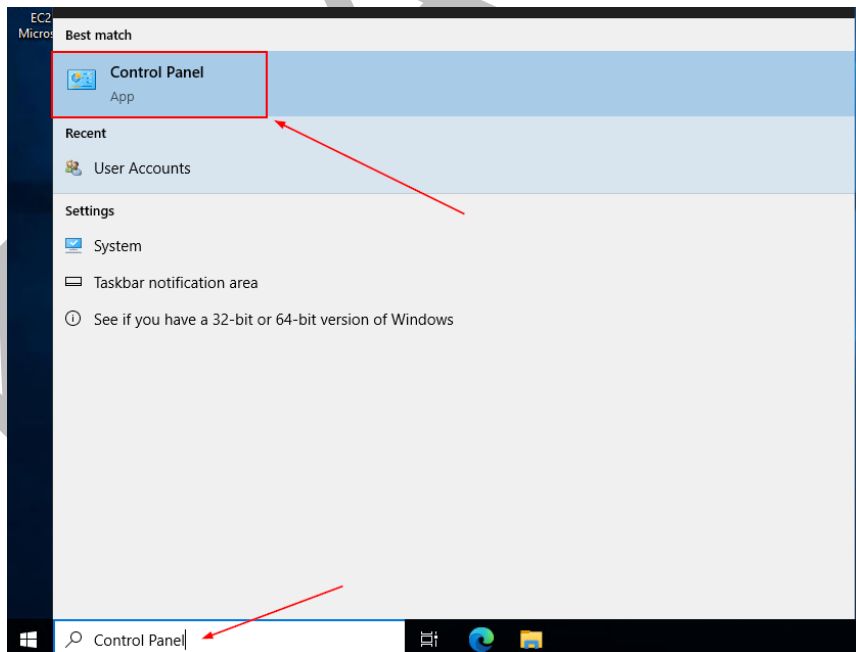
Resetting the password for a Windows instance in AWS can be a quick and easy way to regain access to the instance or improve its security posture. By modifying the user data script to include a new password for the Administrator account, you can ensure that you can log in to the instance using the new password.

In this section we will be providing detailed steps to reset the password of the Windows Instance

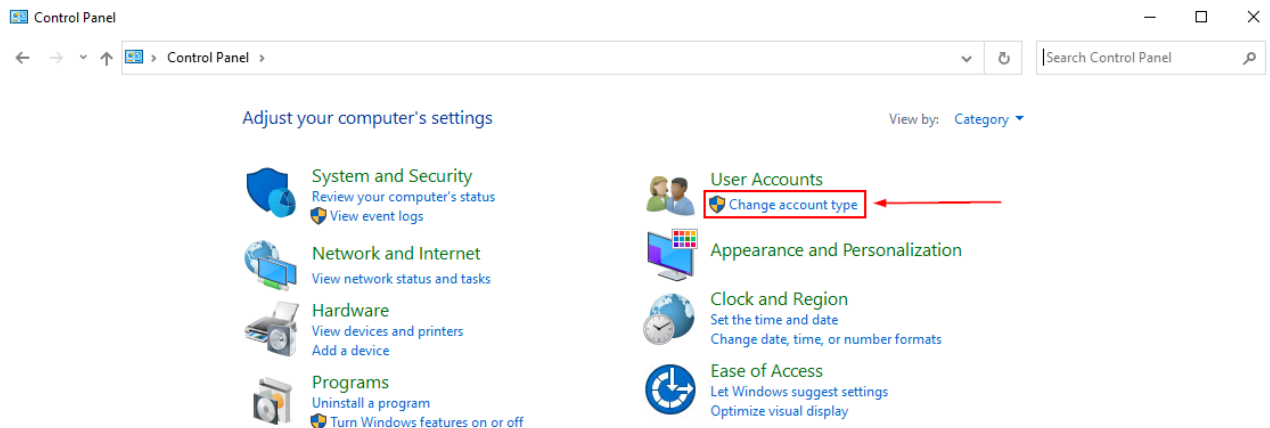
1. First you need to log in to the windows Instance.



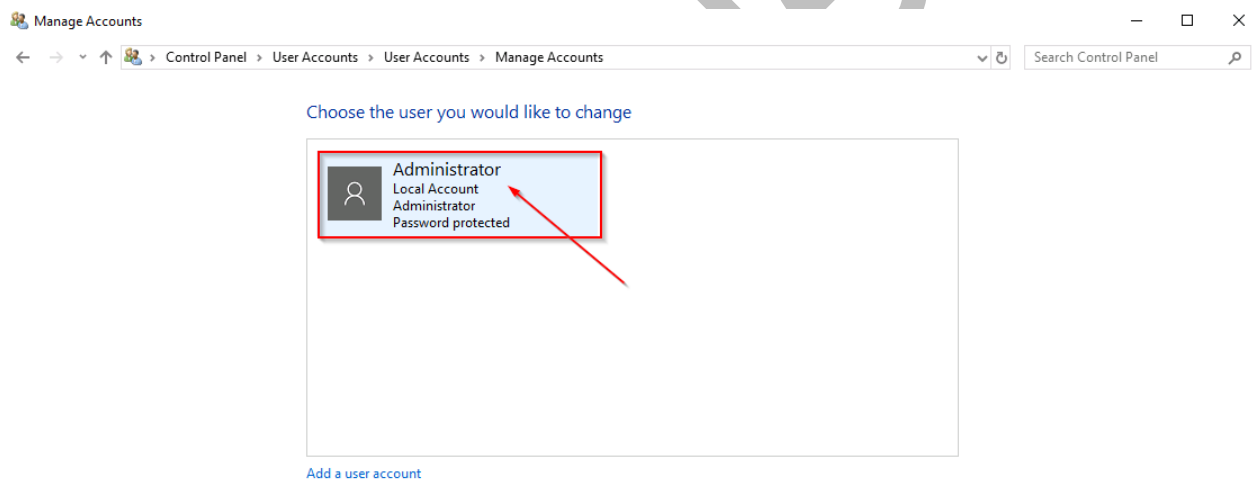
2. In search box at bottom of windows search "Control Panel". Open the **Control Panel**.



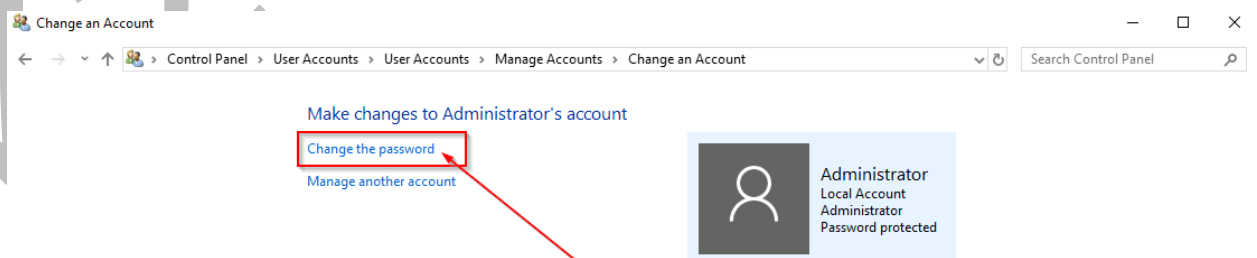
3. Under **User Accounts**, click on **Change Account Type**



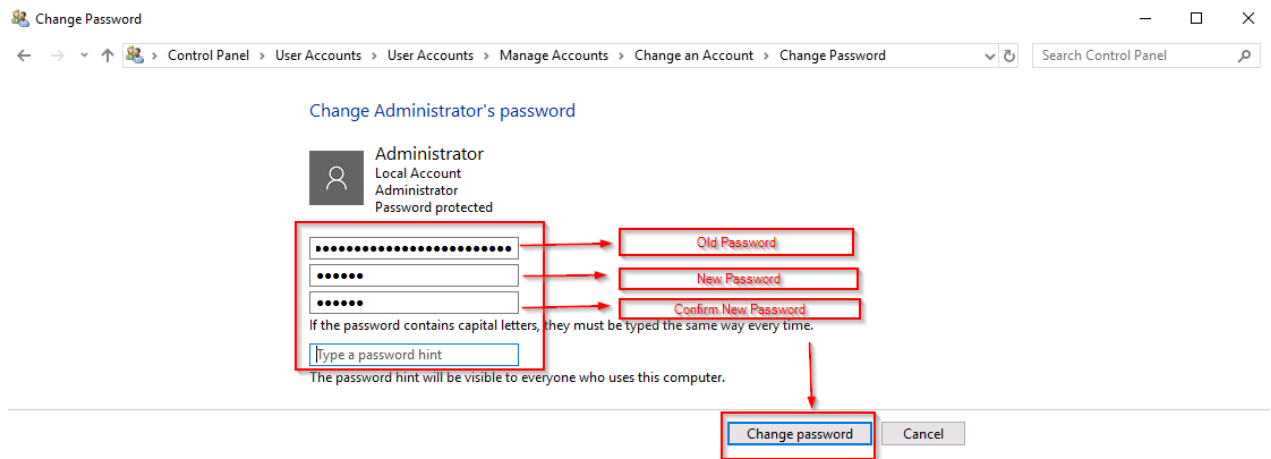
4. Click on the **Administrator** account.



5. Click on **Change Password**.

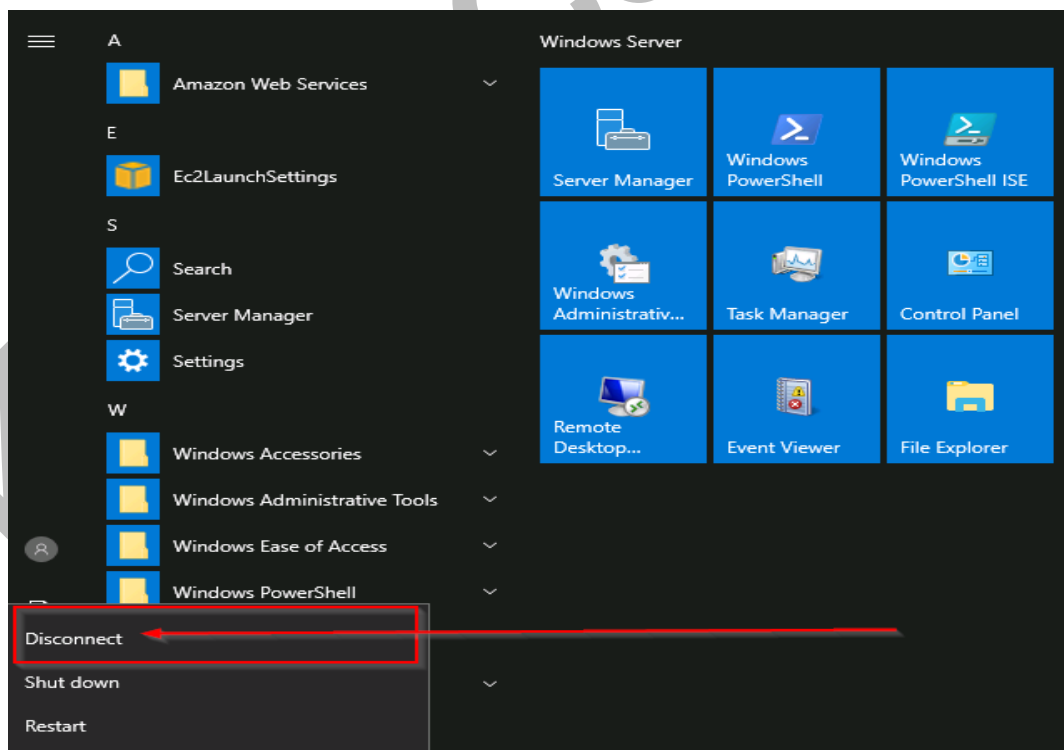


6. In order to change the password, first **type the old password of the Windows Instance**, then **new password** and then finally **confirm the new password**. Then click on **Change password**.

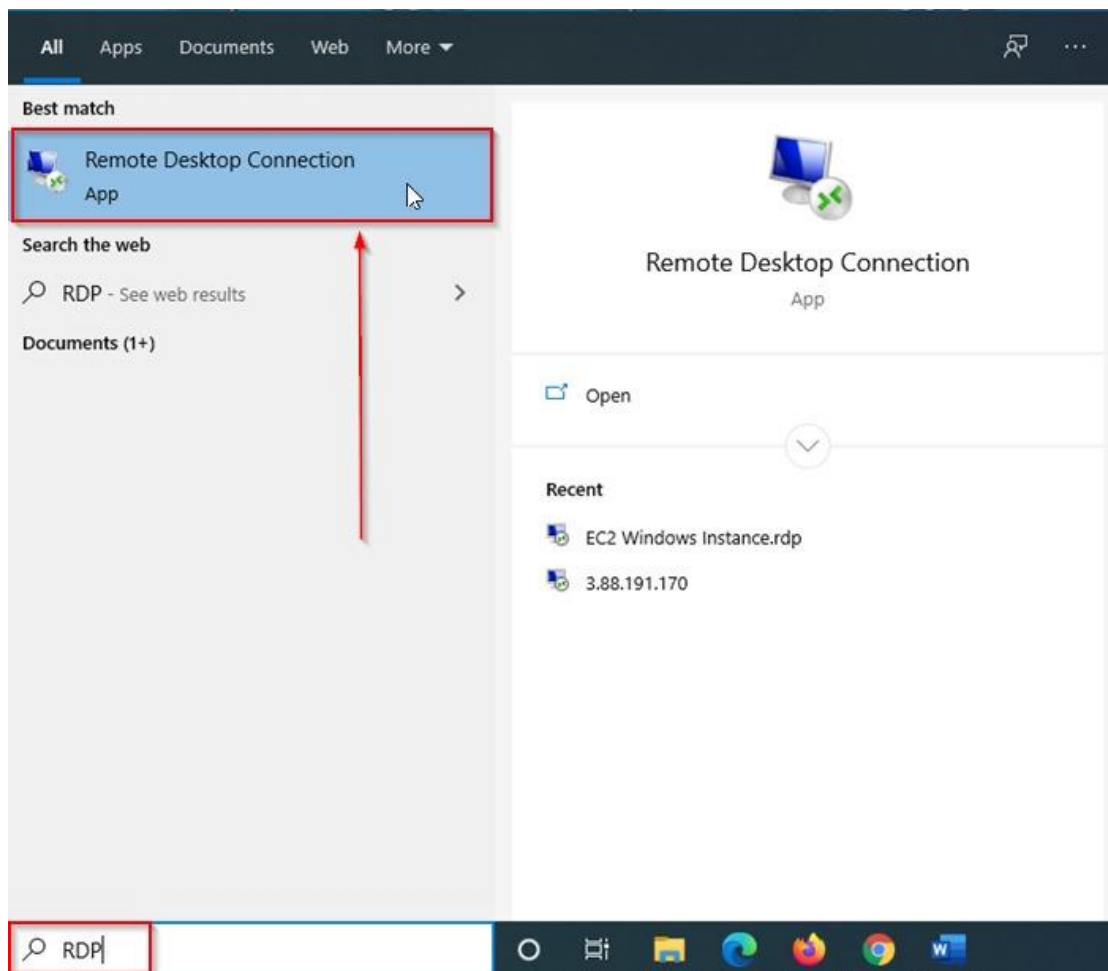


Note: For **old password**, you have to give the password of the **Windows Instance** with the help of which you logged in to the instance. Then you have to type a **new password** and **confirm new password** by typing it again. You can type any password of your choice for the new password. You can also type a hint for your password if you want.

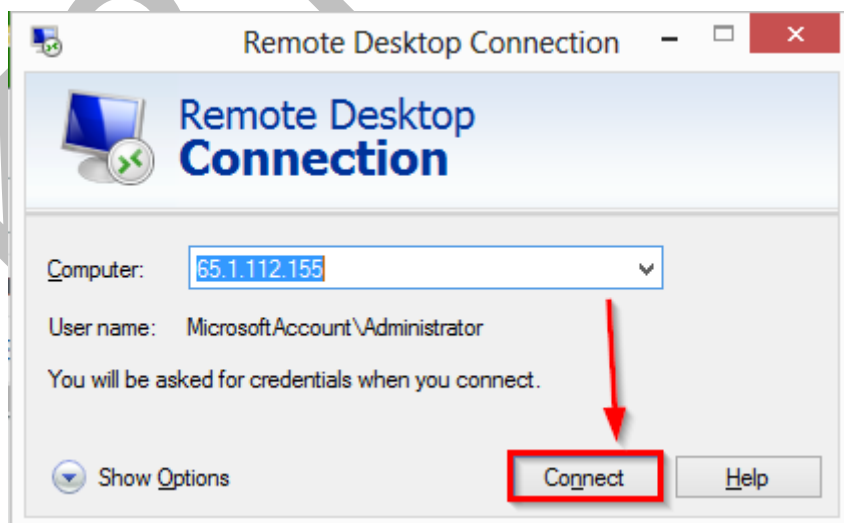
7. Now disconnect from the Windows Instance. By clicking on **Start button > Power > Disconnect**.



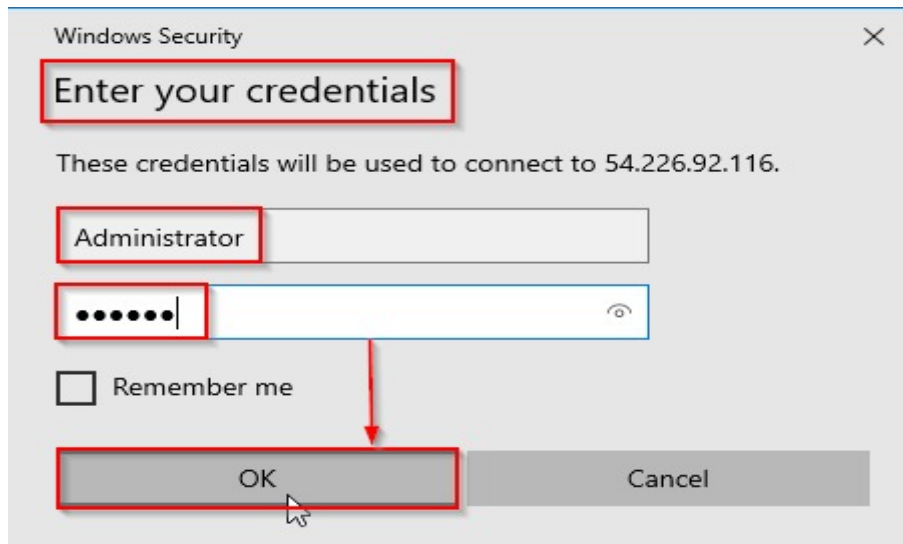
8. Now in order to connect to the instance again, search for **RDP** in the Start menu of your laptop and open it.



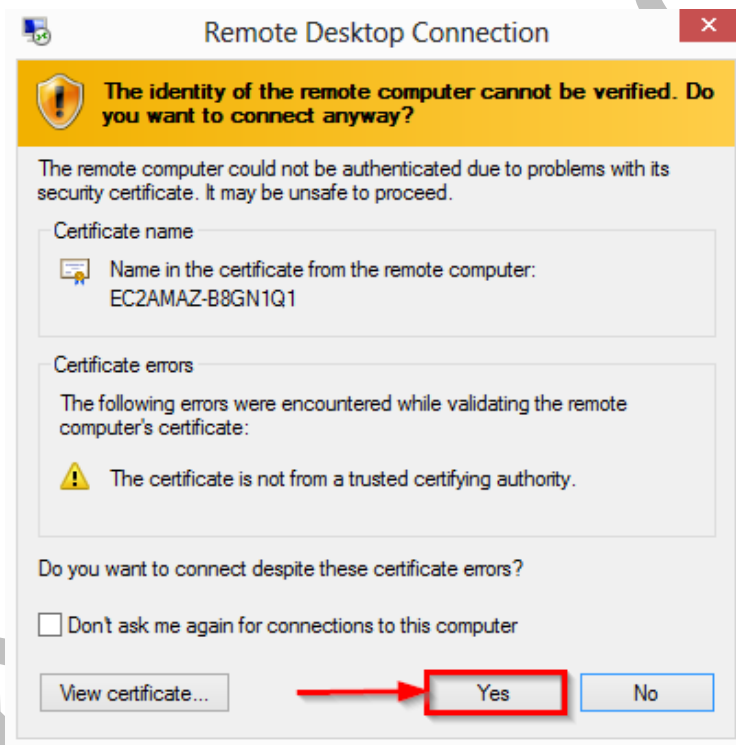
9. Now copy the **Public IP** of the Windows Instance and paste it here and click on **Connect**



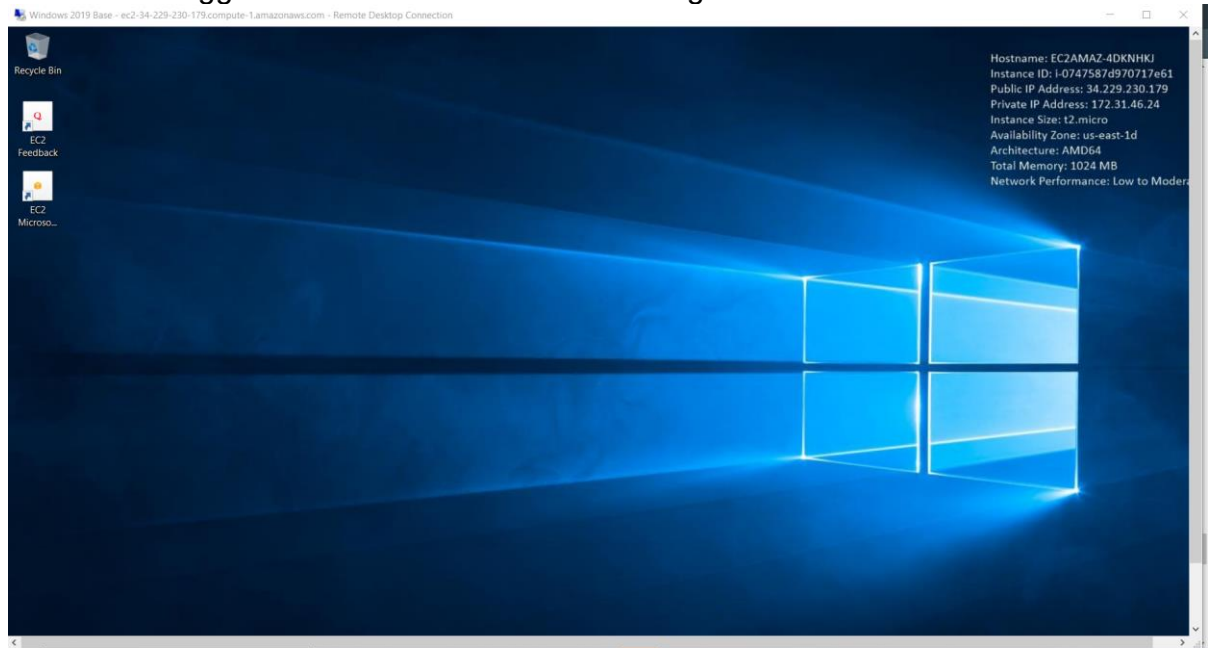
10. Now enter the new password that you had reset.



11. Click on **Yes**.



12. You will be logged in to the Windows Instance again.

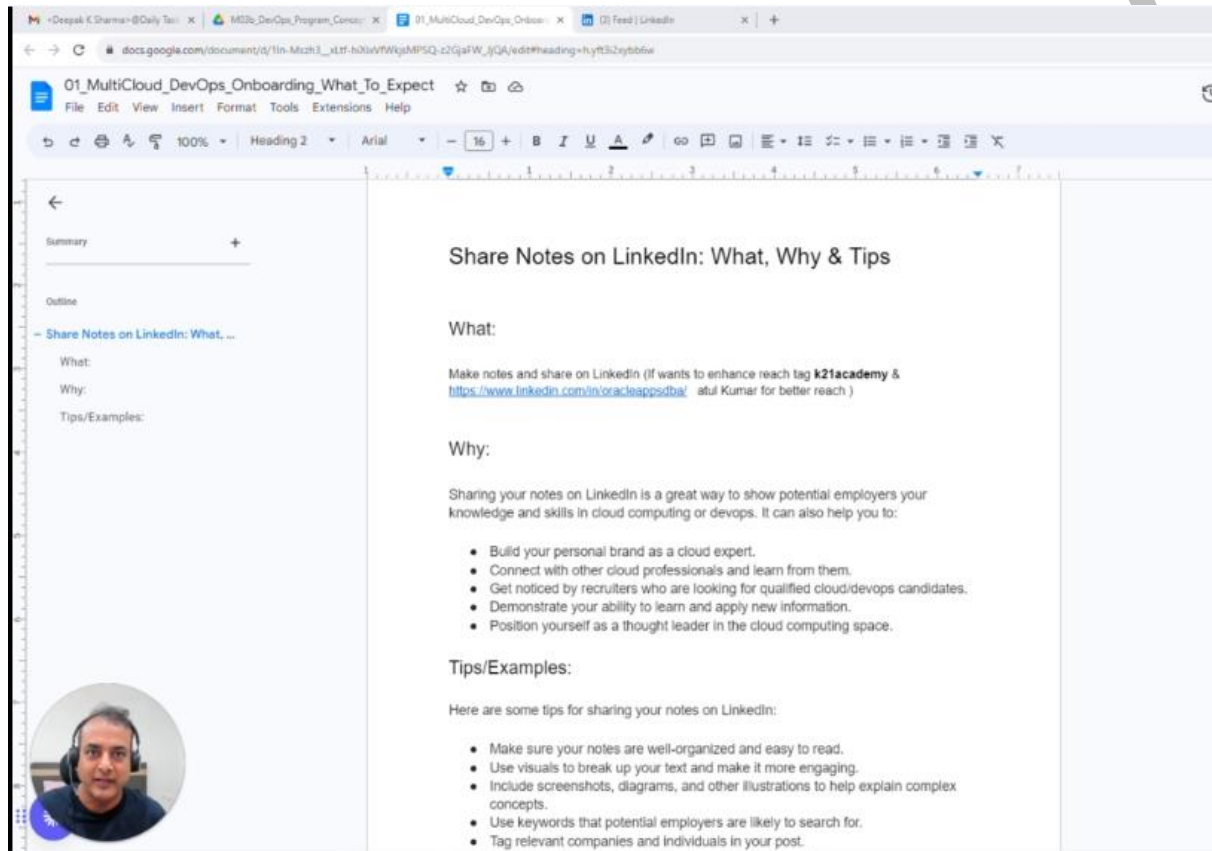


We have successfully reset the password of the Windows Instance and logged back into the Instance using the new password

7 SHARE YOUR LEARNINGS

In this section, you are going to share whatever you learned during this lab on LinkedIn and WhatsApp

 <https://www.loom.com/share/d91f9abbec794f4f85e7d99bb82c2ff6>



- As you know, building a strong professional profile is essential in today's job market. One way to showcase your skills and knowledge is by sharing your labs and projects on LinkedIn.
- Your LinkedIn profile is a powerful tool in your job search. Sharing your labs and projects is just one way to demonstrate your expertise and stand out to potential employers. If you don't have LinkedIn, we strongly recommend you create one for yourself.
- By doing so, you can demonstrate to potential employers or connections that you have hands-on experience in your field and are actively engaged in learning and growing your skills.

7.1 On LinkedIn

Take a screenshot of lab and share it on your LinkedIn. So, this will attract recruiters and employers on your profile and increase reach. Do remember to tag **K21Academy**, as we will circulate in our network too, to increase your reach.

Please refer these following examples on how to share on LinkedIn (don't limit yourself to just these type). Share learnings in points, notes 📝 or anything that can help others.

Here is a sample which includes steps inside a post followed by screenshot:

Sumti Mehta • 1st
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2w • Edited • 🌐

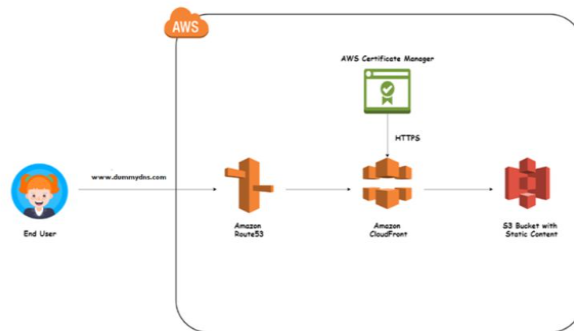
Hi There,

Today I have Provisioned an SSL/TLS Certificates using AWS Certificate Manager. It was fun and learned a lot of services integrated in the deployment.

To do this deployment I had to gone through the following steps:

1. Created an S3 bucket and enabled Static Website Hosting
2. Created a Hosted Zone and Mapped DNS Using Route53
3. Requested SSL/TLS certificates using AWS Certificate Manager
4. And finally, Configured CloudFront Distribution

Thanks, [K21Academy: Learn Cloud From Experts](#) For providing me with the required resources. [#aws](#) [#awscloud](#) [#awsservices](#) [#learning](#) [#cloud](#) [#cloud](#) [#project](#) [#learning](#)



Sumti Mehta and 15 others

Sahid Anwar Hussain • 1st
DevOps Engineer & Cloud Support at K21Academy: Learn Cloud From Experts
2w • 🌐

Hi There,

Today I have deployed an awesome tool to monitor an entire K8s cluster. It was fun and learned a lot diring the deployment.

To do this deployment I had to gone through the following steps:

1. Created an multinode K8s cluster.
2. Created a namespace named "monitoring".
3. Created clusterRole & clusterRole Binding for Prometheus.
4. Created the ConfigMap enclosing all the necessary Data for deploying Prometheus
5. Created a Deployment file for Prometheus
6. Finally I ran all the files and wallah.. Its done.

Thanks, [K21Academy: Learn Cloud From Experts](#) For providing me with the required resources. [#kubernetes](#) [#k8s](#) [#devops](#) [#learning](#) [#cloud](#) [#cloud](#) [#project](#) [#learning](#) [#prometheus](#) [#monitoring](#)

The screenshot shows the Prometheus web interface with the 'Targets' page selected. It displays a table of monitored targets, including kube-state-metrics, kubernetes-api-servers, kubernetes-cadvisor, and kubernetes-metrics.

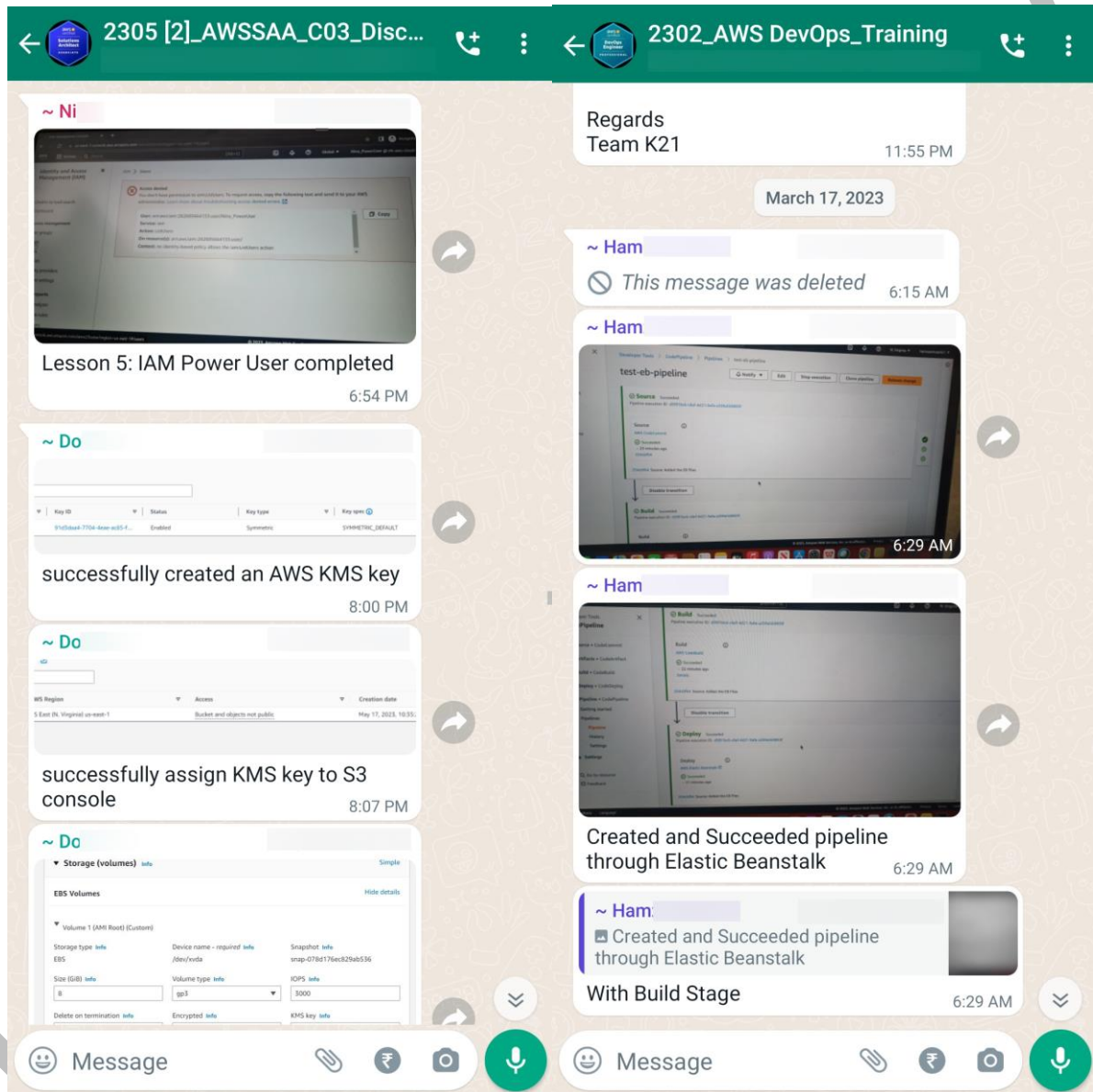
Endpoint	Status	Labels	Last Scrape	Scrape Duration	Error
http://kubernetes-master:8080/metrics	DOWN	namespace=kube-system,service=kube-system,task=kube-system	3.969s ago	12.422ms	Get "http://kubernetes-master:8080/metrics": dial tcp 10.0.0.1:8080: connect: connection refused
https://10.0.0.1:8080/metrics	UP	namespace=kube-system,service=kube-system,task=kube-system	3.969s ago	95.479ms	
https://10.0.0.1:8080/metrics	UP	namespace=kube-system,service=kube-system,task=kube-system	3.969s ago	95.479ms	
https://10.0.0.1:8080/metrics	UP	namespace=kube-system,service=kube-system,task=kube-system	3.969s ago	95.479ms	

Suraj . and 12 others

7.2 On WhatsApp

Share screenshot of lab in WhatsApp group so this will encourage others and also boost your confidence on the progressive path you are following.

Also, we will keep track of your progress on labs so do share your screenshot in WhatsApp. group too.



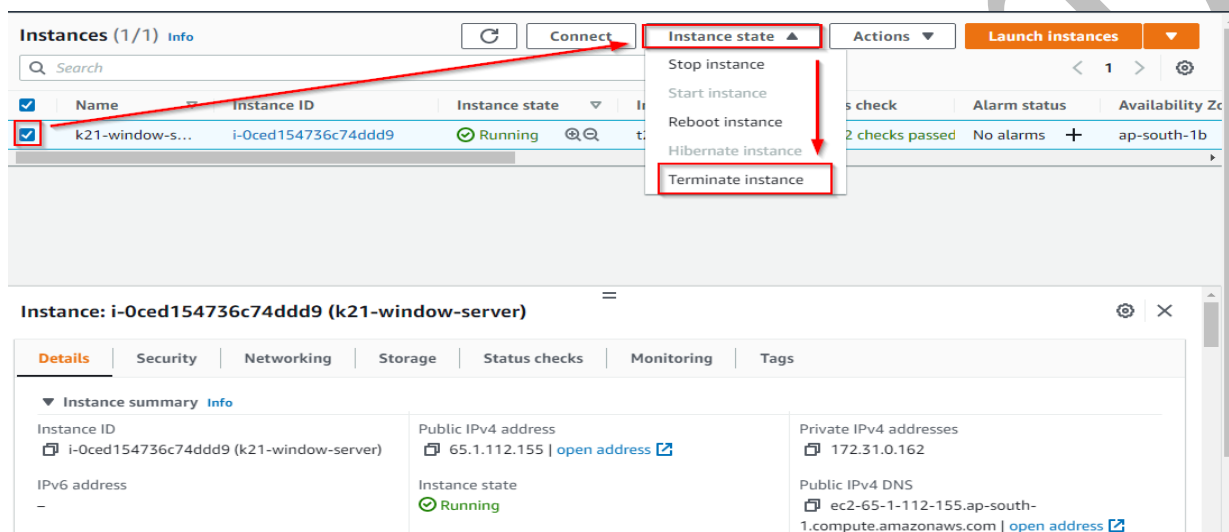
8 DELETING/ CLEANUP

Note: In this section we will be removing all the resources created in this lab so that we can avoid any unnecessary charges.

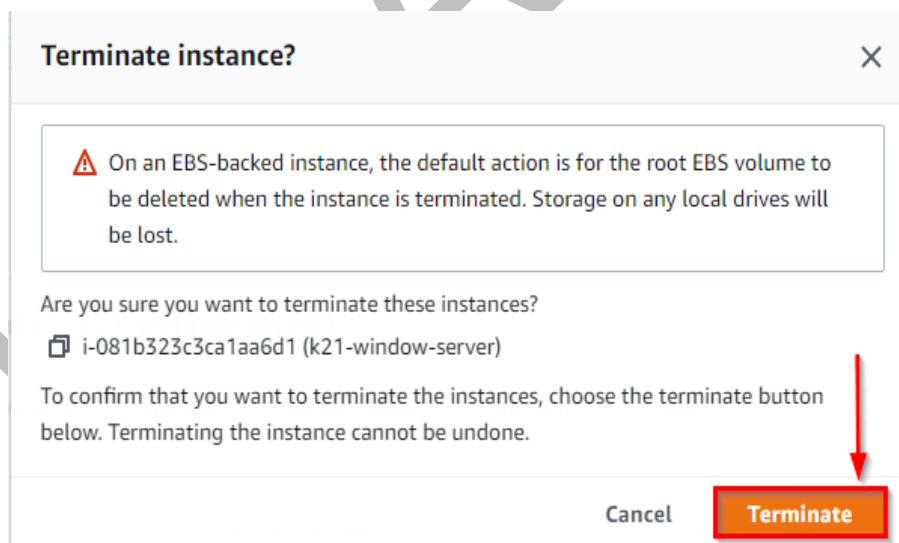
8.1 Terminating the Windows Instance

Note: This section will cover steps to terminate the windows instance

Go to your Instance and select the Windows Instance. Click on **Instance State** and click on **Terminate**



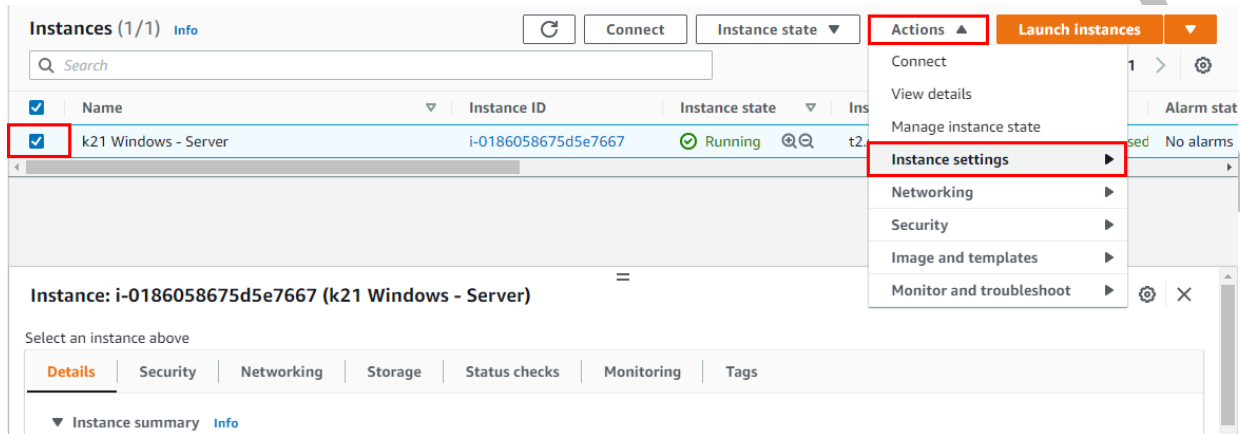
1. Click on **Terminate** to terminate the Instance



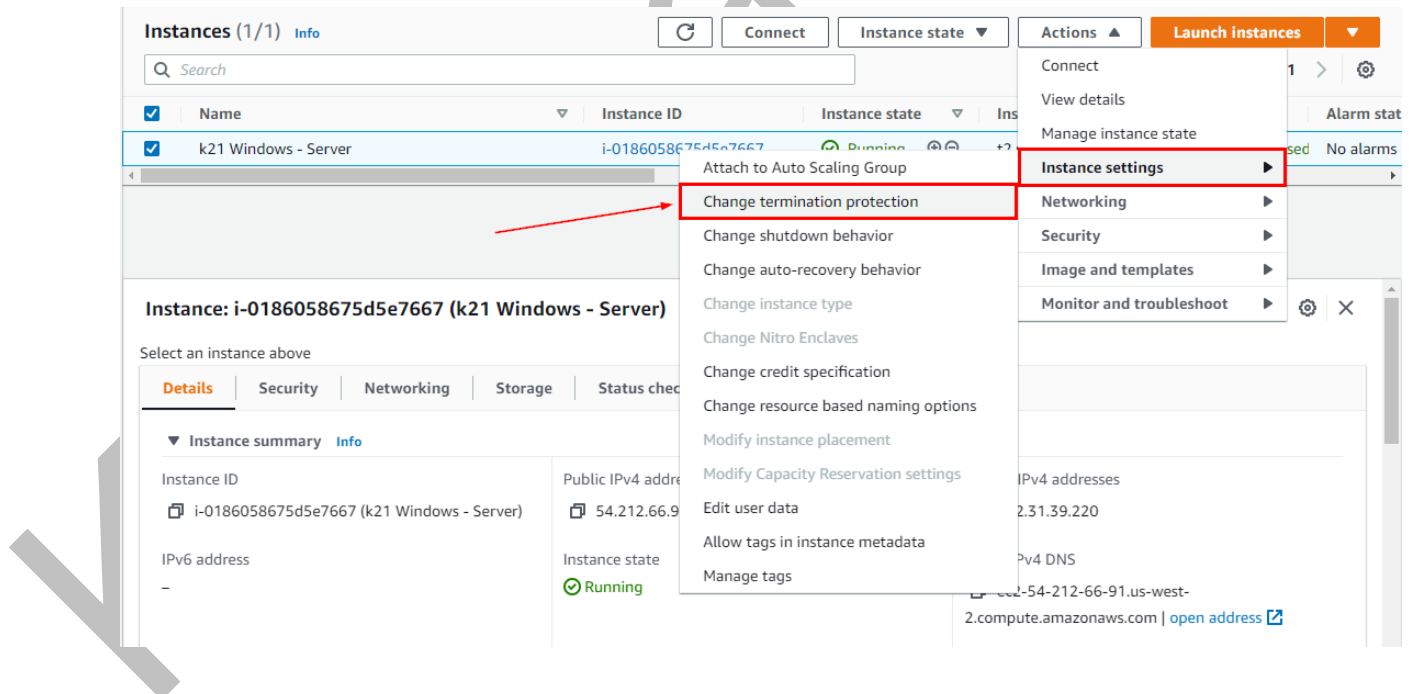
8.2 Termination Prevention of Instance

Note: To protect against the data loss caused by accidental termination of an Amazon EC2 instance, we need enable the termination protection and if you want to disable Instance termination protection you can follow these steps.

1. Select the **Instance**, click on **Action** and then click on the **Instance settings**



- a. Now click on **Change Termination Protection**



2. While creating the Instance its already enabled, for disabling it **Uncheck the button** and click on **Save**

EC2 > Instances > i-0186058675d5e7667 > Change termination protection

Change termination protection [Info](#)

Enable termination protection to prevent your instance from being accidentally terminated.

Instance ID
i-0186058675d5e7667 (k21 Windows - Server)

Termination protection
☒ Enable

Cancel **Save**

9 TROUBLESHOOTING SECTION

In this section, we have mentioned commonly occurred errors you might get while performing this lab.

9.1 Not Able to Connect to RDP

9.1.1 ISSUE

Sometime we may face this kind of issue which may be due to one of the reasons mentioned in the screenshot

Remote Desktop Connection



Remote Desktop can't connect to the remote computer for one of these reasons:

- 1) Remote access to the server is not enabled
- 2) The remote computer is turned off
- 3) The remote computer is not available on the network

Make sure the remote computer is turned on and connected to the network, and that remote access is enabled.

OK

Help

9.1.2 CAUSE

This error is caused due to your laptop that it blocking the outgoing traffic of port 3389, there may be a Window defender or the laptop you are using is may be your office laptop so they don't allow the traffic to go outside.

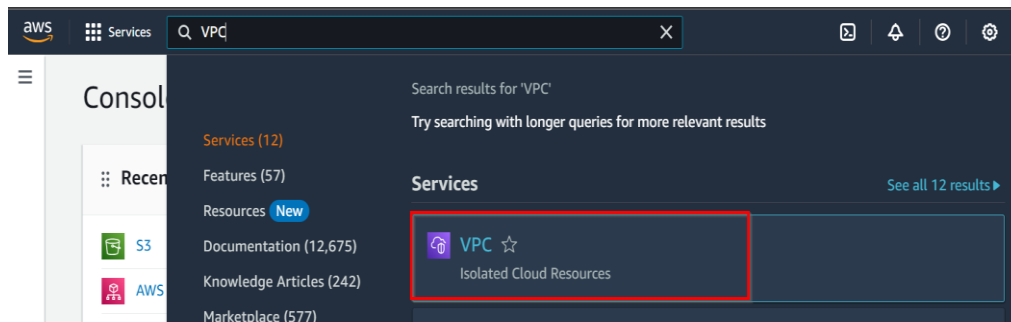
9.1.3 FIX

Your firewall is preventing the traffic from outgoing port, check your firewall and enable RDP on it. Make sure your computer has remote access and you have proper network access.

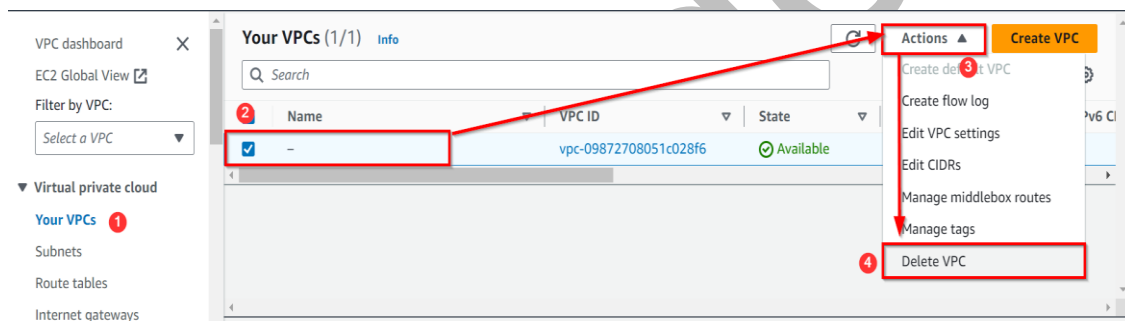
OR

To Fix this error

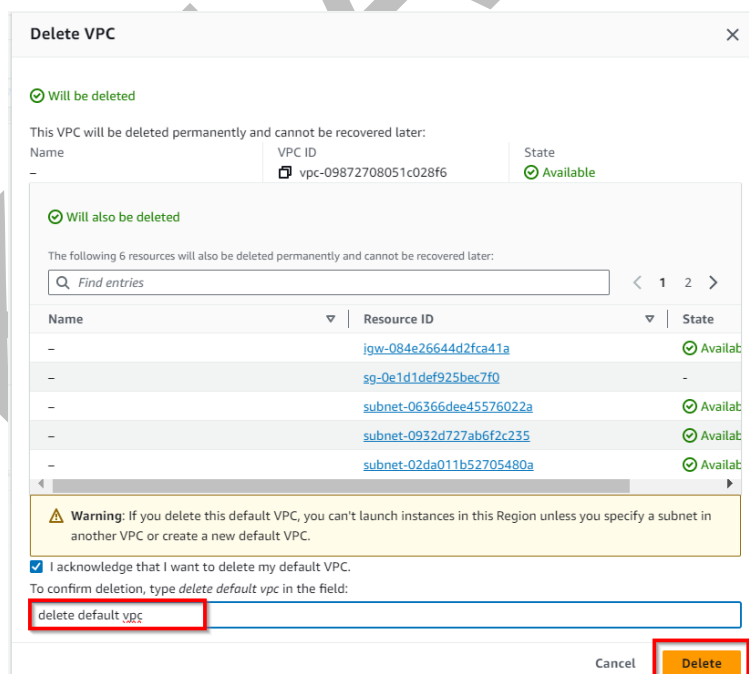
1. Check the default Internet Gateway or Create the Internet Gateway and attach it to your VPC.
or
2. Reconfigure the Default VPC.
 - Search VPC in search bar and select it.



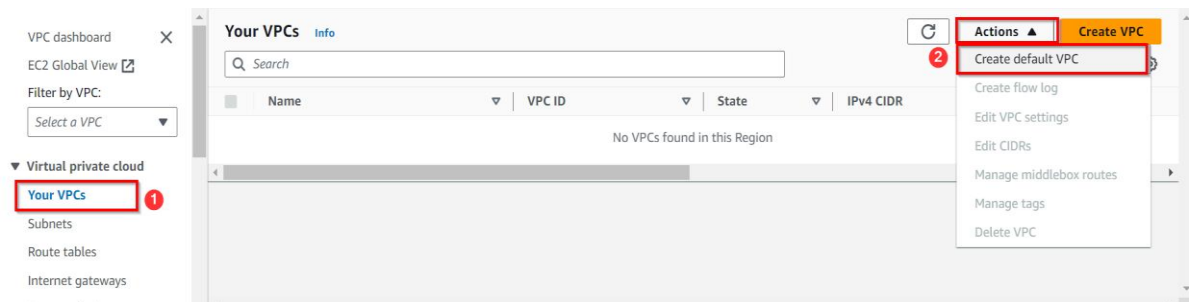
- Delete the existing default VPC.



- Now confirm the deletion

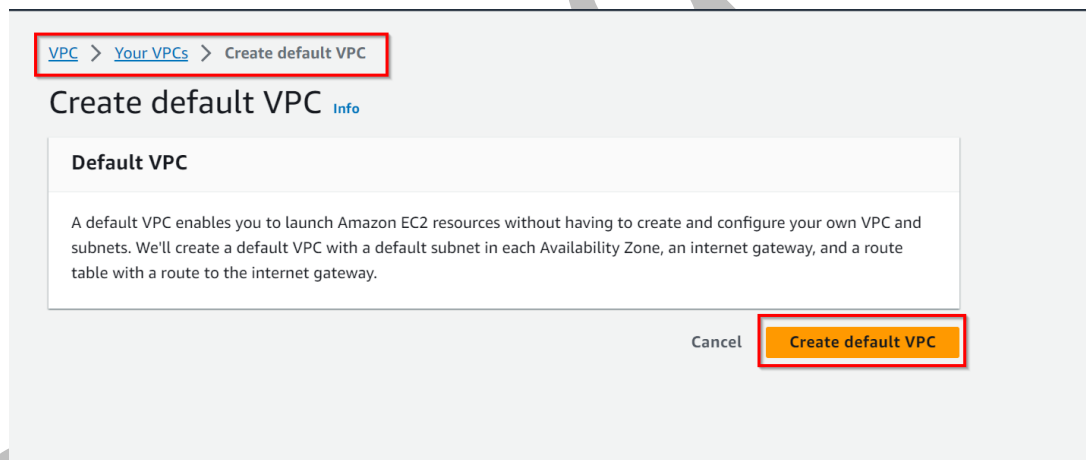


- Go to Action.
- Under the Action, click on create Default VPC.



A default VPC enables you to launch Amazon EC2 resources without having to create and configure your own VPC and subnets. We'll create a default VPC with a default subnet in each Availability Zone, **an internet gateway**, and a **route table** with a route to the internet gateway.

- Now Click on create default VPC.



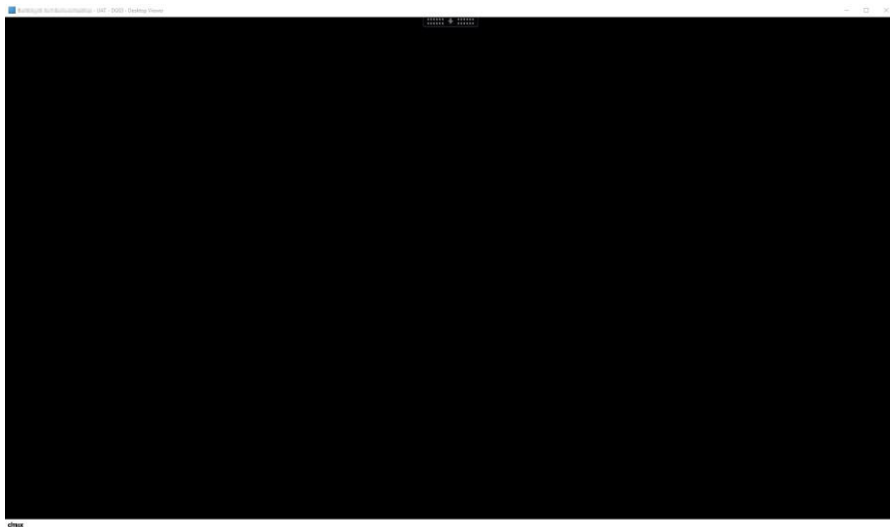
Now you will able to Create & connect to the EC2.

Suggestion: Try to create & connect the EC2 in another region as well.

9.2 RDP displays a Black Screen

9.2.1 ISSUE

Sometime we may face this kind of issue that RDP displays a black screen instead of the desktop.



9.2.2 CAUSE

This error is caused due to RDP old Version or you are not logged in as Administrator or server is over-utilized.

9.2.3 FIX

Try the following to resolve this issue:

- 1) Check the console output for additional information. To get the console output for your instance using the Amazon EC2 console, select the instance, and then choose Actions, Monitor and troubleshoot, Get system log.
- 2) Verify that you are running the latest version of your RDP client.
- 3) If the server is running a full-screen application, it might have stopped responding. Use Ctrl+Shift+Esc to start Windows Task Manager, and then close the application.
- 4) If the server is over-utilized, it might have stopped responding. To monitor the instance using the Amazon EC2 console, select the instance and then select the Monitoring tab.
- 5) If you are using Remote Desktop Connection, try starting it with the /admin option as follows.

```
mstsc /v:instance /admin
```

9.3 Lost Key Pair

9.3.1 ISSUE

If your key pair is lost (.pem file) and unable to login into instance.

9.3.2 CAUSE

You might have deleted your key pair.

9.3.3 FIX

Create an AMI and launch a new instance

When you use EC2Config or EC2Launch to reset a lost password, you must use its key pair to retrieve the administrator password. If you've lost the key pair, you can create an AMI of the existing instance, and then launch a new instance. You can then select a new key pair by following the instance launch wizard. Follow these steps:

- 1) Create a new key pair, and then save the private key file in a safe place. You can create a key pair using the console.

Note: To give the new key pair the same name as the lost key pair, you must first delete the lost key pair.

- 2) From the Amazon EC2 console, choose **Instances** from the navigation pane.
- 3) Select your instance. From the **Description** tab, note the **Instance type**, **VPC ID**, **Subnet ID**, **Security groups**, and **IAM role** for the instance.
- 4) Stop your instance. **Warning:** If this instance has an instance store volume, any data on it is lost when the instance is stopped. If the instance shutdown behavior is set to **Terminate**, the instance terminates when it is stopped.
- 5) Select your instance. For **Actions**, choose **Image**, **Create Image**.
For **Image name**, enter a name.
- 6) Choose **Create Image**, and then choose **Close**.
- 7) Choose **AMIs** from the navigation pane. If the **Status** is **pending**, the AMI is still being created. When the **Status** is **available**, continue to the next step.
- 8) Select the AMI, and then choose **Launch**.
- 9) Complete the wizard. Be sure to select the **same Instance type**, **VPC ID**, **Subnet ID**, **Security groups**, and **IAM role** as the instance that you are replacing.

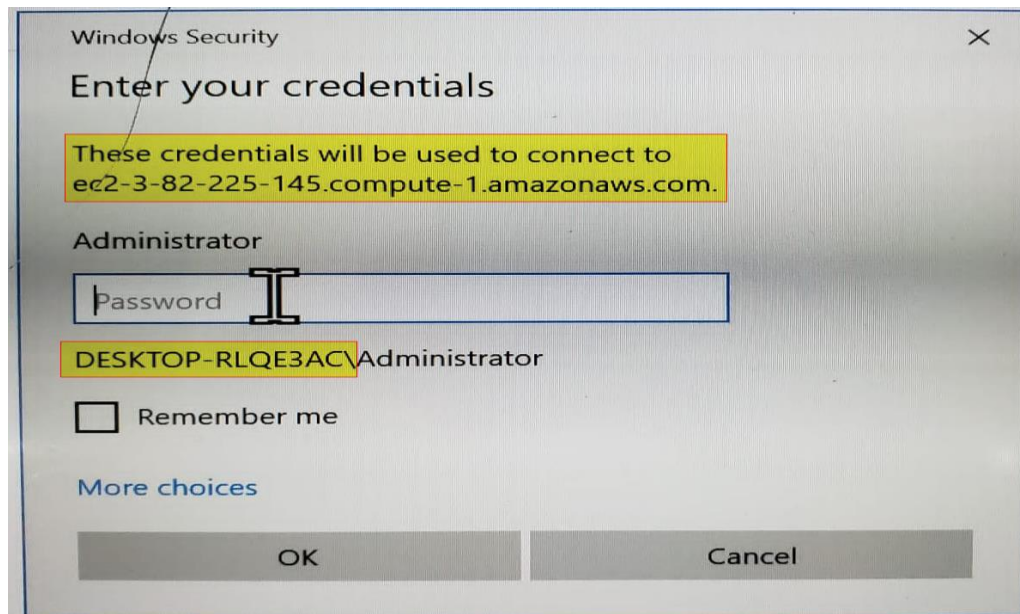
For **Select a key pair**, choose the new key pair.

- 10) Now that the private key is replaced, you can reset the administrator password.

9.4 Unable to login after changing password

9.4.1 ISSUE

Not able to login to the window instance after changing the password.



9.4.2 CAUSE

You are using the wrong credentials or IP address/DNS name.

9.4.3 FIX

When connecting to window EC2 after changing the password, you need to open RDP from your Laptop or Desktop not from the virtual machine. Also, the username provided must be as **Administrator** only.

10 SUMMARY

This activity guide covers the steps to:

1. Creating a Windows EC2 Instance
 - a. Launching the Windows Instance
 - b. Connecting to the Windows Instance
2. Deleting/Cleanup
 - a. Stopping the Windows Instance
 - b. Terminating the Windows Instance
 - c. Termination Prevention of Instance